

TmFeO₃ Fe–Tm K^- Coupling

Restored derivation: physical J operator,
PJP projection, local SU(3) qutrit basis, transported-gauge bond formulas, and the
 physical cross-peak mechanism (vertices, conservation laws, Liouville pathways), grounded
 in measured TmFeO₃ parameters

Working note

June 2026

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1 Purpose and conceptual reset

The goal of this note is to formulate the Fe–Tm coupling in a way that is mathematically well-defined all the way down to the projected qutrit operators. The central object is the K^- coupling, namely the bilinear coupling between an Fe spin and the time-reversal-odd Tm qutrit operators

$$\lambda_j^- = \begin{pmatrix} \lambda_j^2 \\ \lambda_j^5 \\ \lambda_j^7 \end{pmatrix}. \quad (1)$$

The superscript “ $-$ ” means time-reversal odd. The notation j labels the Tm sublattice, not a spatial vector component.

Key point

There are three different mathematical objects that must not be confused:

1. $\mathbf{S}_i$: an Fe spin vector, stored in global crystallographic axes.
2. $\mathbf{J}_j$: the physical Tm angular momentum operator in the full $J = 6$ Hilbert space.
3. λ_j^a : a coordinate operator in the local three-level qutrit basis at Tm sublattice j .

Only $\mathbf{S}$ and the full $\mathbf{J}$ are axial vectors. A Gell-Mann operator λ_j^a is not a spatial vector component. It transforms by the adjoint action induced by the chosen qutrit bases.

The physically supported route is therefore

$$\text{physical } \mathbf{S} \text{ and } \mathbf{J} \longrightarrow H_{\text{Fe-Tm}} = \mathbf{S}^T \mathbf{A} \mathbf{J} \longrightarrow P_j \mathbf{J} P_j \longrightarrow K_{ij}^- \lambda_j^- . \quad (2)$$

One should not start by assuming that the space-group rotation $R_x(\pi)$ acts as a closed onsite SU(3) rotation in one fixed Tm qutrit. That is precisely the dangerous shortcut.

2 Physical system: verified TmFeO₃ parameters

Before any formal manipulation we fix the physical setting and the numbers the model is calibrated against. All values below are experimental properties of bulk TmFeO₃; the simulation parameters sit close to them, and the small residual differences are stated explicitly so that “model” and “measurement” are never silently conflated.

Structure and sites. TmFeO₃ is an orthorhombically distorted perovskite, space group $Pbnm$ (No. 62, a nonstandard setting of $Pnma$), with $Z = 4$ formula units per cell. The Fe³⁺ ions occupy the Wyckoff $4b$ sites, whose site symmetry is $\bar{1}$: *each Fe sits on an inversion centre*, so spatial inversion maps every Fe sublattice to itself. The Tm³⁺ ions occupy the $4c$ sites $(x, y, \frac{1}{4})$, whose site symmetry is the single mirror m (the plane $z = \frac{1}{4}$, i.e. m_z); inversion therefore exchanges Tm sublattices in pairs. This is exactly the Fe/Tm permutation structure used in the Pbnm-ingredients section, and it is dictated by the crystallography, not chosen. In particular, the local mirror $m_z = S_1^{-1}I$ that generates the λ^5, λ^7 sign is the physical $4c$ site mirror.

Magnetic ions. Fe³⁺ is $3d^5$, a spin-only ${}^6S_{5/2}$ ion ($S = 5/2$, $L \approx 0$, $g \approx 2$), represented classically by the unit vector $\mathbf{S}_i$. Tm³⁺ is $4f^{12}$ with ground multiplet 3H_6 ($J = 6$, 13 states). In the $C_s(m)$ crystal field the multiplet splits into 13 *singlets* (a non-Kramers ion in a site without rotational degeneracy has no protected doublets); the qutrit keeps the three lowest, $|E_1\rangle, |E_2\rangle, |E_3\rangle$, and represents them by local SU(3) Gell-Mann operators.

Magnetic order and the Γ_2 ground state. The Fe sublattice orders G-type antiferromagnetically at $T_N \approx 632$ K, with a Dzyaloshinskii–Moriya canting that produces weak ferromagnetism. In Bertaut/Yamaguchi mode notation [5] the relevant collinear-mode multiplets are

$$\Gamma_2 = (F_x, C_y, G_z), \quad \Gamma_4 = (G_x, A_y, F_z). \quad (3)$$

On cooling, TmFeO₃ undergoes a continuous spin reorientation $\Gamma_4 \rightarrow \Gamma_{24} \rightarrow \Gamma_2$ between $T_2 \approx 85$ K and $T_1 \approx 93$ K [3, 6]. Below ~ 85 K the equilibrium is Γ_2 : the main AFM vector $\mathbf{G}$ lies along c (G_z) and the weak ferromagnetic moment $\mathbf{F}$ along a (F_x). This $F_x + G_z$ state is the “baseline” used throughout the numerical sections.

Excitation energies: the two axes of the cross peak. The two THz excitations that define the target cross peak are an Fe magnon and a Tm crystal-field transition:

Excitation	THz	meV	cm ⁻¹	role
Fe quasi-FM magnon	0.28	1.16	9.3	low Fe mode
Fe quasi-AFM magnon	0.86	3.56	28.7	pumped q_{AFM} (ω_τ axis)
Tm CEF E_{12} (R_1)	0.52	2.15	17.3	detected coherence (ω_t axis)
Tm CEF E_{23} (R_2)	0.61	2.52	20.3	sets Δ_{13} with E_{12}
Tm CEF E_{13} (R_3)	1.15	4.76	38.4	third qutrit level

Conversion factor: $1 \text{ THz} = 4.136 \text{ meV} = 33.36 \text{ cm}^{-1} = 47.99 \text{ K}$. The three CEF lines are mutually consistent with a three-singlet qutrit, $E_{12} + E_{23} \approx E_{13}$ ($0.52 + 0.61 \approx 1.13 \text{ THz}$), so the onsite splittings in $H_{\text{Tm},0}$ are $\Delta_{12} \approx 0.52$ and $\Delta_{13} \approx 1.13 \text{ THz}$. The simulation uses $q_{\text{AFM}} \approx 0.90$ and $E_{12} \approx 0.50 \text{ THz}$, within a few percent of the measured 0.86 and 0.52 THz [3, 4]. The cross-peak question is whether a coherence on the $\omega_\tau = q_{\text{AFM}}$ axis can be converted into a coherence on the $\omega_t = E_{12}$ axis; the two differ by $\Delta \approx 0.34 \text{ THz}$, a number we return to under energy conservation in section 16.

3 Pbnm ingredients

We use the generators

$$S_1 : (x, y, z) \mapsto (-x, -y, z + \frac{1}{2}), \quad (4)$$

$$S_2 : (x, y, z) \mapsto (x + \frac{1}{2}, -y + \frac{1}{2}, -z), \quad (5)$$

$$I : (x, y, z) \mapsto (-x, -y, -z). \quad (6)$$

For axial vectors, the corresponding spin-space matrices are

$$R_{S_1} = \text{diag}(-1, -1, +1) \equiv R_z, \quad R_{S_2} = \text{diag}(+1, -1, -1) \equiv R_x, \quad R_I = \mathbf{1}. \quad (7)$$

Their product gives

$$R_{S_1 S_2} = R_y = \text{diag}(-1, +1, -1). \quad (8)$$

The sublattice permutations are

Operation	Fe sublattice action	Tm sublattice action
S_1	(03)(12)	(03)(12)
S_2	(01)(23)	(01)(23)
$S_1 S_2$	(02)(13)	(02)(13)
I	identity	(03)(12)

Thus inversion fixes each Fe sublattice but pairs $\text{Tm}_0 \leftrightarrow \text{Tm}_3$ and $\text{Tm}_1 \leftrightarrow \text{Tm}_2$.

4 Full physical Fe–Tm exchange before projection

Let a Fe–Tm bond be denoted

$$b = (i, j; \mathbf{n}), \quad (9)$$

where i is an Fe sublattice, j is a Tm sublattice, and $\mathbf{n}$ is the Tm-cell offset relative to the Fe cell. Before any qutrit truncation, the most general bilinear physical exchange is

$$H_b^{\text{phys}} = \mathbf{S}_i^T A_b \mathbf{J}_j. \quad (10)$$

Here A_b is a real 3×3 tensor with rows labelled by Fe spin components and columns labelled by Tm angular-momentum components:

$$A_b = \begin{pmatrix} A_{xx} & A_{xy} & A_{xz} \\ A_{yx} & A_{yy} & A_{yz} \\ A_{zx} & A_{zy} & A_{zz} \end{pmatrix}_b. \quad (11)$$

Under a space-group operation g , the physical axial vectors transform as

$$\mathbf{S}_i \mapsto R_g \mathbf{S}_{\sigma_g^{\text{Fe}}(i)}, \quad \mathbf{J}_j \mapsto R_g \mathbf{J}_{\sigma_g^{\text{Tm}}(j)}. \quad (12)$$

Therefore invariance of eq. (10) requires

$$A_{gb} = R_g^T A_b R_g. \quad (13)$$

This equation is safe because it is written in the full physical Hilbert space. No $SU(3)$ truncation has been used.

Warning

The statement that $\mathbf{J}$ is an axial vector is a statement about the full $J = 6$ angular-momentum operator. It is not a statement that $(\lambda^2, \lambda^5, \lambda^7)$ form a spatial vector.

5 Local qutrit bases and the PJP projection

At each Tm sublattice j , define the local qutrit projector

$$P_j = \sum_{n=1}^3 |E_{n,j}\rangle \langle E_{n,j}|. \quad (14)$$

The local qutrit basis is

$$\mathcal{B}_j = \{|E_{1,j}\rangle, |E_{2,j}\rangle, |E_{3,j}\rangle\}. \quad (15)$$

The Gell-Mann operator at that Tm site is defined by

$$\lambda_j^a = \sum_{m,n=1}^3 |E_{m,j}\rangle (\lambda^a)_{mn} \langle E_{n,j}|. \quad (16)$$

Thus λ_j^a is a local operator coordinate in the Tm qutrit. It is not a component of a spatial vector.

The physical projected angular momentum is

$$\tilde{J}_{j,\alpha} = P_j J_\alpha P_j. \quad (17)$$

In the local qutrit basis it is expanded as

$$P_j J_\alpha P_j = \sum_{a \in \{2,5,7\}} \mu_{j,\alpha a} \lambda_j^a. \quad (18)$$

Lemma: why only $\lambda^2, \lambda^5, \lambda^7$ appear

The restriction of (18) to the three *antisymmetric* Gell-Mann matrices is exact, not a small-multipole approximation, and it rests on one physical fact specific to this system. Tm^{3+} is a non-Kramers $4f^{12}$ ion at a site of symmetry m , so its CEF eigenstates are nondegenerate singlets and can be chosen *real* (time-reversal symmetric). In any real basis the angular-momentum operator J_α is purely imaginary and antisymmetric, hence $\langle E_m | J_\alpha | E_n \rangle$ is imaginary and antisymmetric. The only imaginary-antisymmetric Gell-Mann matrices are $\lambda^2, \lambda^5, \lambda^7$, so $P_j J_\alpha P_j$ has *no* component on $\lambda^{1,3,4,6,8}$. The time-odd (λ^-) versus time-even (λ^+) split used throughout this note is therefore a rigorous consequence of time reversal acting on a real singlet basis, not a modelling choice.

For a reference local basis, the projected moment matrix is

$$\mu_0 = \begin{pmatrix} 0 & 2.3915 & 0.9128 \\ 0 & -2.7866 & 0.4655 \\ 5.264 & 0 & 0 \end{pmatrix}, \quad \text{columns } (2, 5, 7). \quad (19)$$

Equivalently,

$$P_0 J_x P_0 = 2.3915\lambda_0^5 + 0.9128\lambda_0^7, \quad (20)$$

$$P_0 J_y P_0 = -2.7866\lambda_0^5 + 0.4655\lambda_0^7, \quad (21)$$

$$P_0 J_z P_0 = 5.264\lambda_0^2. \quad (22)$$

The physical magnetic moment is

$$\mathbf{m}_j^{\text{Tm}} = -g_J \mu_B P_j \mathbf{J} P_j. \quad (23)$$

Thus a lab magnetic field couples to the projected moment as

$$H_{B,\text{Tm}}(t) = -g_J \mu_B \sum_{j,\alpha,a} B_\alpha(t) \mu_{j,\alpha a} \lambda_j^a. \quad (24)$$

The coefficient $\mu_{j,\alpha a}$ carries all local-basis information.

6 The actual SU(3) symmetry action: W_g and D_g

Let g map Tm sublattice j to $j' = \sigma_g^{\text{Tm}}(j)$. The exact qutrit-basis map is

$$[W_g^{(j)}]_{mn} = \langle E_{m,j'} | U_g | E_{n,j} \rangle. \quad (25)$$

It acts on qutrit states as

$$U_g | E_{n,j} \rangle = \sum_m | E_{m,j'} \rangle [W_g^{(j)}]_{mn}. \quad (26)$$

It induces an adjoint action on qutrit operators:

$$U_g \lambda_j^a U_g^\dagger = \sum_b \left[D_g^{(j)} \right]_{ba} \lambda_{j'}^b, \quad (27)$$

where

$$\left[D_g^{(j)} \right]_{ba} = \frac{1}{2} \text{Tr} \left[\lambda^b W_g^{(j)} \lambda^a W_g^{(j)\dagger} \right]. \quad (28)$$

Key point

$W_g^{(j)}$ acts on qutrit states. $D_g^{(j)}$ acts on qutrit operators. R_g acts on physical spatial vector components. These are different matrices in different spaces.

The projected moment covariance is

$$U_g (P_j J_\alpha P_j) U_g^\dagger = \sum_\beta R_{g,\alpha\beta} P_{j'} J_\beta P_{j'}. \quad (29)$$

Substituting eqs. (18) and (27) gives

$$\sum_a \mu_{j,\alpha a} \left[D_g^{(j)} \right]_{ba} = \sum_\beta R_{g,\alpha\beta} \mu_{j',\beta b}. \quad (30)$$

In matrix notation, if D is defined by $\lambda^a \mapsto D_{ba} \lambda^b$, then

$$\mu_{j'} = R_g^T \mu_j D_g^{(j)T}. \quad (31)$$

This is the more theoretically supported relation. It replaces the unsafe shortcut $\lambda_{\text{global}} = \mu^{-1} R \mu \lambda_{\text{local}}$.

Warning

The matrix $\mu^{-1}R_x\mu$ is generally not the adjoint representation of a unitary qutrit rotation. The operation S_2 maps one Tm sublattice to another; it need not act as a closed onsite operation in one fixed qutrit. The remedy is to work with $W_g^{(j)}$ and $D_g^{(j)}$, or to choose transported bases that make selected $D_g^{(j)}$ trivial by construction.

7 Transported gauge: a practical but explicit convention

Choose Tm_0 as the reference qutrit. Define

$$\mathcal{B}_0 = \{|E_{1,0}\rangle, |E_{2,0}\rangle, |E_{3,0}\rangle\}, \quad (32)$$

$$\mathcal{B}_1 = U_{S_2}\mathcal{B}_0, \quad (33)$$

$$\mathcal{B}_2 = U_{S_1S_2}\mathcal{B}_0, \quad (34)$$

$$\mathcal{B}_3 = U_{S_1}\mathcal{B}_0. \quad (35)$$

This is the transported gauge. In this gauge,

$$D_{S_2}^{(0)} = D_{S_1S_2}^{(0)} = D_{S_1}^{(0)} = \mathbf{1}. \quad (36)$$

After restricting to the corresponding transported maps, the proper-coset qutrit action is therefore absorbed into the basis definition. This does not mean that R_x acts trivially on the qutrit. It means we chose the target qutrit basis to be the image of the source qutrit basis.

The physical global projected moment on site j is then

$$\boxed{P_j J_\alpha^{\text{global}} P_j = \sum_{\beta,a} (R_j)_{\alpha\beta} \mu_{0,\beta a} \lambda_j^a}, \quad (37)$$

where

$$R_0 = \text{diag}(+, +, +), \quad R_1 = \text{diag}(+, -, -), \quad R_2 = \text{diag}(-, +, -), \quad R_3 = \text{diag}(-, -, +). \quad (38)$$

This is how a global lab field talks to local Tm λ_j operators in transported gauge:

$$H_{B,\text{Tm}}(t) = -g_J \mu_B \sum_{j,a} \left[\sum_{\alpha,\beta} B_\alpha(t) (R_j)_{\alpha\beta} \mu_{0,\beta a} \right] \lambda_j^a. \quad (39)$$

8 Local mirror and the origin of the λ^5, λ^7 sign

Two operations map Tm_0 to Tm_3 : S_1 and I . In transported gauge the Tm_3 basis is defined with S_1 . Therefore the inversion map differs from the transported map by the local operation

$$h = S_1^{-1}I. \quad (40)$$

Using the coordinate maps,

$$h : (x, y, z) \mapsto (x, y, -z + \frac{1}{2}), \quad (41)$$

which is a local mirror m_z up to a translation.

For a J_z basis state, the mirror can be written as $m_z = IC_{2z}$. Since C_{2z} acts as $e^{-i\pi J_z}$ and the $4f^{12}$ parity is common to all states,

$$U(m_z)|J, m\rangle = (-1)^m |J, m\rangle \quad (42)$$

after dropping the common parity phase. The first two CEF states live in the even- m sector, while the third lives in the odd- m sector. Hence

$$U(m_z)|E_1\rangle = +|E_1\rangle, \quad U(m_z)|E_2\rangle = +|E_2\rangle, \quad U(m_z)|E_3\rangle = -|E_3\rangle. \quad (43)$$

Thus in the qutrit basis

$$W_{m_z} = \text{diag}(+1, +1, -1). \quad (44)$$

Conjugating the Gell-Mann operators gives

$$\lambda^2 \sim -i|E_1\rangle\langle E_2| + i|E_2\rangle\langle E_1| \Rightarrow +\lambda^2, \quad (45)$$

$$\lambda^5 \sim -i|E_1\rangle\langle E_3| + i|E_3\rangle\langle E_1| \Rightarrow -\lambda^5, \quad (46)$$

$$\lambda^7 \sim -i|E_2\rangle\langle E_3| + i|E_3\rangle\langle E_2| \Rightarrow -\lambda^7. \quad (47)$$

Therefore

$$\boxed{M_- = \text{diag}(+1, -1, -1) \quad \text{on} \quad (\lambda^2, \lambda^5, \lambda^7).} \quad (48)$$

9 From physical exchange to projected K^-

Project eq. (10) to the Tm qutrit:

$$H_b^{\text{phys}} = \sum_{\alpha\beta} S_{i,\alpha}(A_b)_{\alpha\beta} P_j J_\beta P_j \quad (49)$$

$$= \sum_{\alpha,a} S_{i,\alpha} \left[\sum_{\beta} (A_b)_{\alpha\beta} \mu_{j,\beta a} \right] \lambda_j^a. \quad (50)$$

Thus

$$\boxed{K_{b,\alpha a}^- = \sum_{\beta} (A_b)_{\alpha\beta} \mu_{j,\beta a}, \quad a \in \{2, 5, 7\}.} \quad (51)$$

In matrix form,

$$\boxed{K_b^- = A_b \mu_j.} \quad (52)$$

The fully gauge-covariant symmetry rule follows by applying both the Fe rotation and the local qutrit adjoint action:

$$H_b^{K^-} = \mathbf{S}_i^T K_b^- \boldsymbol{\lambda}_j^-. \quad (53)$$

Under g ,

$$\mathbf{S}_i \mapsto R_g \mathbf{S}_{\sigma_g^{\text{Fe}}(i)}, \quad \lambda_j^a \mapsto \sum_b \left[D_g^{(j)} \right]_{ba} \lambda_{\sigma_g^{\text{Tm}}(j)}^b. \quad (54)$$

Therefore

$$\boxed{K_{gb,\beta b}^- = \sum_{\alpha,a} R_{g,\alpha\beta} K_{b,\alpha a}^- \left[D_g^{(j)} \right]_{ba}.} \quad (55)$$

This index equation is the safest form. In matrix notation, with the same D_{ba} convention,

$$\boxed{K_{gb}^- = R_g^T K_b^- D_g^{(j)T}.} \quad (56)$$

Implementation prescription

Equation (55) is the implementation anchor. The Fe row index is transformed by the spatial axial-vector matrix R_g . The Tm λ column index is transformed by the qutrit adjoint matrix $D_g^{(j)}$. In transported gauge many proper-coset D matrices are absorbed into the basis, but

the Fe row signs remain if Fe spins are stored in global axes.

10 One K^- orbit in transported gauge

Take a reference bond with coefficient

$$K_0^- = \begin{pmatrix} k_{x2} & k_{x5} & k_{x7} \\ k_{y2} & k_{y5} & k_{y7} \\ k_{z2} & k_{z5} & k_{z7} \end{pmatrix} = (\mathbf{k}_2 \quad \mathbf{k}_5 \quad \mathbf{k}_7). \quad (57)$$

For the direct proper-coset partners in transported gauge,

$$K_{ii}^- = R_i K_0^-. \quad (58)$$

For the inversion-related partners,

$$K_{i,3-i}^- = R_i K_0^- M_-. \quad (59)$$

Thus the orbit Hamiltonian is

$$H_{K^-}^{\text{orbit}} = \sum_{i=0}^3 \mathbf{S}_i^T R_i [\mathbf{k}_2(\lambda_i^2 + \lambda_{3-i}^2) + \mathbf{k}_5(\lambda_i^5 - \lambda_{3-i}^5) + \mathbf{k}_7(\lambda_i^7 - \lambda_{3-i}^7)]. \quad (60)$$

This equation is often the most useful way to read the selection rule:

$$\lambda^2 \text{ couples to the inversion-even Tm combination,} \quad (61)$$

$$\lambda^5, \lambda^7 \text{ couple to inversion-odd Tm combinations.} \quad (62)$$

It does not mean that the mirror-odd terms are identically zero. They vanish only after projecting onto an inversion-even Tm sector.

10.1 All eight coefficients of one orbit

Bond	Generating operation	Coefficient	Meaning
(Fe ₀ , Tm ₀)	E	$K_{00}^- = K_0^-$	reference
(Fe ₁ , Tm ₁)	S_2	$K_{11}^- = R_x K_0^-$	Fe y, z rows flip
(Fe ₂ , Tm ₂)	$S_1 S_2$	$K_{22}^- = R_y K_0^-$	Fe x, z rows flip
(Fe ₃ , Tm ₃)	S_1	$K_{33}^- = R_z K_0^-$	Fe x, y rows flip
(Fe ₀ , Tm ₃)	I	$K_{03}^- = K_0^- M_-$	λ^5, λ^7 columns flip
(Fe ₁ , Tm ₂)	$S_2 I$	$K_{12}^- = R_x K_0^- M_-$	row flip + odd columns
(Fe ₂ , Tm ₁)	$S_1 S_2 I$	$K_{21}^- = R_y K_0^- M_-$	row flip + odd columns
(Fe ₃ , Tm ₀)	$S_1 I$	$K_{30}^- = R_z K_0^- M_-$	row flip + odd columns

11 Fixed-Fe and fixed-Tm viewpoints

For a fixed Fe sublattice i , the mirror-odd part is

$$H_i^{5,7} = \mathbf{S}_i^T R_i \mathbf{k}_5 (\lambda_i^5 - \lambda_{3-i}^5) + \mathbf{S}_i^T R_i \mathbf{k}_7 (\lambda_i^7 - \lambda_{3-i}^7). \quad (63)$$

So the cancellation is in the Tm sector: if $\lambda_i^{5,7} = \lambda_{3-i}^{5,7}$, then this fixed-Fe contribution vanishes.

For a fixed Tm site j , collect the λ_j^5 terms:

$$H_{\lambda_j^5} = (\mathbf{S}_j^T R_j \mathbf{k}_5 - \mathbf{S}_{3-j}^T R_{3-j} \mathbf{k}_5) \lambda_j^5. \quad (64)$$

Similarly,

$$H_{\lambda_j^7} = (\mathbf{S}_j^T R_j \mathbf{k}_7 - \mathbf{S}_{3-j}^T R_{3-j} \mathbf{k}_7) \lambda_j^7. \quad (65)$$

This is not $\mathbf{S}_j - \mathbf{S}_{I(j)}$, because inversion fixes the Fe sublattice. The index $3 - j$ enters because it is the Fe partner of the other member of this particular bond orbit.

12 Including the Fe and Tm Hamiltonians

A minimal Fe Hamiltonian in global axes is

$$H_{\text{Fe}} = \sum_{\langle mn \rangle_1} [J_{1xy}(S_m^x S_n^x + S_m^y S_n^y) + J_{1z} S_m^z S_n^z] \quad (66)$$

$$+ \sum_{\langle mn \rangle_2} [J_{2xy}(S_m^x S_n^x + S_m^y S_n^y) + J_{2z} S_m^z S_n^z] \quad (67)$$

$$+ \sum_m [D_x (S_m^x)^2 + D_y (S_m^y)^2 + D_z (S_m^z)^2]. \quad (68)$$

The onsite Tm qutrit Hamiltonian with $E_1 = 0$, $E_2 = \Delta_{12}$, $E_3 = \Delta_{13}$ is

$$H_{\text{Tm},0} = \sum_j \left[-\frac{\Delta_{12}}{2} \lambda_j^3 + \frac{\Delta_{12} - 2\Delta_{13}}{2\sqrt{3}} \lambda_j^8 \right] + \text{constant}. \quad (69)$$

The full leading model restricted to K^- is

$$H = H_{\text{Fe}} + H_{\text{Tm},0} + \sum_{\nu} H_{K^-, \nu} + H_{B,\text{Fe}} + H_{B,\text{Tm}} + H_{E,\text{Tm}}. \quad (70)$$

Here ν labels inequivalent Fe–Tm geometric bond orbits. Each $H_{K^-, \nu}$ has the form of eq. (60) but with its own reference columns $\mathbf{k}_{\nu,2}$, $\mathbf{k}_{\nu,5}$, $\mathbf{k}_{\nu,7}$ and its own cell offsets.

13 Higher-order symmetry-allowed couplings involving external fields and two Fe spins

The previous sections derived the lowest-order Fe–Tm bilinear

$$\mathbf{S}_i^T A_b \mathbf{J}_j \longrightarrow \mathbf{S}_i^T K_b^- \boldsymbol{\lambda}_j^-. \quad (71)$$

The same logic gives the higher-order couplings. The key rule is that one must first write a physical operator with well-defined spatial and time-reversal transformation properties, and only then project the Tm operator into the local SU(3) qutrit space.

We will use two Tm qutrit sectors:

$$\boldsymbol{\lambda}_j^- = (\lambda_j^2, \lambda_j^5, \lambda_j^7)^T, \quad \text{time-reversal odd}, \quad (72)$$

$$\boldsymbol{\lambda}_j^+ = (\lambda_j^1, \lambda_j^3, \lambda_j^4, \lambda_j^6, \lambda_j^8)^T, \quad \text{time-reversal even}. \quad (73)$$

Often only the off-diagonal time-even coherences are kept,

$$\boldsymbol{\lambda}_{j,\text{off}}^+ = (\lambda_j^1, \lambda_j^4, \lambda_j^6)^T. \quad (74)$$

The local mirror m_z acts on the three CEF states as

$$W_{m_z} = \text{diag}(+1, +1, -1). \quad (75)$$

Therefore the mirror/parity matrices in the local qutrit basis are

$$M_- = \text{diag}(+1, -1, -1), \quad \text{basis } (\lambda^2, \lambda^5, \lambda^7), \quad (76)$$

$$M_+ = \text{diag}(+1, +1, -1, -1, +1), \quad \text{basis } (\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8), \quad (77)$$

$$M_{+,\text{off}} = \text{diag}(+1, -1, -1), \quad \text{basis } (\lambda^1, \lambda^4, \lambda^6). \quad (78)$$

Key point

Time reversal fixes which qutrit sector can appear:

Factor	$\mathcal{T}$ parity	Inversion type	Comment
$\mathbf{S}$	odd	axial, even under I	Fe spin
$\mathbf{J}$	odd	axial, even under I	physical Tm angular momentum
$\mathbf{B}$	odd	axial, even under I	magnetic field
$\mathbf{E}$	even	polar, odd under I	electric field
λ^-	odd	M_-	$\lambda^2, \lambda^5, \lambda^7$
λ^+	even	M_+	$\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8$

A term in the Hamiltonian must be time-reversal even unless the external nonequilibrium drive is being treated as an explicitly time-dependent source. Thus ESJ uses the λ^- sector after PJP projection, while $BS\lambda$ and $SS\lambda$ naturally use the λ^+ sector.

13.1 Electric-field-assisted Fe–Tm term: $E + S + J$

Start from a physical third-order coupling on a bond $b = (i, j; \mathbf{n})$,

$$H_b^{ESJ}(t) = \sum_{\eta, \alpha, \beta} E_\eta(t) S_{i, \alpha} C_{b; \eta \alpha \beta}^{EJ} J_{j, \beta}. \quad (79)$$

Here η is a polar-vector electric-field index, α is an Fe-spin axial-vector index, and β is a physical Tm angular-momentum axial-vector index. Since E is time-even while S and J are time-odd, the product ESJ is time-even.

Now project the physical J operator:

$$P_j J_\beta P_j = \sum_{a \in \{2, 5, 7\}} \mu_{j, \beta a} \lambda_j^a. \quad (80)$$

Then

$$P_j H_b^{ESJ} P_j = \sum_{\eta, \alpha, a \in \{2, 5, 7\}} E_\eta(t) S_{i, \alpha} K_{b; \eta \alpha a}^{E, -} \lambda_j^a, \quad (81)$$

with

$$K_{b; \eta \alpha a}^{E, -} = \sum_{\beta} C_{b; \eta \alpha \beta}^{EJ} \mu_{j, \beta a}. \quad (82)$$

Thus $E + S + J$ is not a new kind of Tm operator; after projection it is an electric-field-assisted source for the same time-odd qutrit operators $\lambda^2, \lambda^5, \lambda^7$.

Gauge-covariant symmetry rule. For a space-group operation g , the polar electric field transforms with the polar matrix Q_g , while S and J transform with the axial matrix R_g . In the projected qutrit basis,

$$U_g \lambda_j^a U_g^\dagger = \sum_b \left[D_g^{(j)} \right]_{ba} \lambda_{\sigma_g(j)}^b. \quad (83)$$

Therefore the projected tensor satisfies the index rule

$$K_{gb; \nu \gamma c}^{E, -} = \sum_{\eta, \alpha, a} Q_{g, \eta \nu} R_{g, \alpha \gamma} K_{b; \eta \alpha a}^{E, -} \left[D_g^{(j)} \right]_{ca}. \quad (84)$$

This is the field-assisted analogue of the K^- covariance rule. The extra Q_g index is the only new ingredient.

Inversion partner. For inversion, $Q_I = -1$ because E is polar, while $R_I = +1$ for axial vectors. In the transported gauge the relative Tm operation gives M_- , so the inversion partner carries

$$\boxed{\text{odd-bond sign matrix for } ESJ : \quad -M_- = \text{diag}(-1, +1, +1).} \quad (85)$$

Thus the selection rule is the opposite of the static K^- term:

$$\lambda^2 : \text{opposite sign on the inversion partner}, \quad (86)$$

$$\lambda^5, \lambda^7 : \text{same sign on the inversion partner}. \quad (87)$$

Transported-gauge orbit formula. Let the reference electric-assisted column vectors be

$$\mathbf{e}_{\eta,2}, \quad \mathbf{e}_{\eta,5}, \quad \mathbf{e}_{\eta,7}, \quad (88)$$

where each is a three-component Fe-spin vector. For the proper coset sending the reference Fe sublattice to i , define Q_i as the polar point matrix and R_i as the axial spin matrix. A compact transported-gauge form is

$$\boxed{H_{\text{orbit}}^{ESJ}(t) = \sum_{i=0}^3 \sum_{\nu} E_{\nu}(t) \mathbf{S}_i^T R_i \sum_{\eta} Q_{i,\eta\nu} \left[\mathbf{e}_{\eta,2}(\lambda_i^2 - \lambda_{3-i}^2) \right.} \quad (89)$$

$$\left. + \mathbf{e}_{\eta,5}(\lambda_i^5 + \lambda_{3-i}^5) + \mathbf{e}_{\eta,7}(\lambda_i^7 + \lambda_{3-i}^7) \right].$$

If the code has already contracted the electric tensor onto a fixed lab polarization, the factor $\sum_{\eta} Q_{i,\eta\nu} E_{\nu}$ should be viewed as a sublattice-dependent effective coefficient. Dropping it is only valid if the coupling parameters have already been fitted separately for each proper-coset partner.

Implementation prescription

For an implementation storing Fe spins in global axes and Tm operators in local qutrit bases, the ESJ odd-bond signs are

$$\lambda^2 : -1, \quad \lambda^5 : +1, \quad \lambda^7 : +1. \quad (90)$$

Additionally, proper-coset partners require the Fe row signs R_i , and field-assisted terms require the electric-field polar-index signs Q_i unless the field polarization has already been folded into orbit-dependent coefficients.

13.2 Magnetic-field-assisted Fe–Tm term: $B + S + \lambda$

A magnetic-field-assisted coupling with one Fe spin and one Tm qutrit operator has the form

$$\boxed{H_b^{BS\lambda}(t) = \sum_{\eta,\alpha,a \in +} B_{\eta}(t) S_{i,\alpha} K_{b;\eta\alpha}^{B,+} \lambda_j^a.} \quad (91)$$

Here $a \in +$ means

$$a \in \{1, 3, 4, 6, 8\}, \quad (92)$$

or, if only off-diagonal coherences are retained,

$$a \in \{1, 4, 6\}. \quad (93)$$

This time-even sector is required because B and S are both time-reversal odd, so BS is time-reversal even.

If one wants to derive this term from a physical Tm time-even operator $\mathcal{O}_\rho^+$, write

$$P_j \mathcal{O}_\rho^+ P_j = o_{j,\rho 0} \mathbf{1} + \sum_{a \in \{1,3,4,6,8\}} o_{j,\rho a} \lambda_j^a. \quad (94)$$

A physical coupling

$$\sum_{\eta, \alpha, \rho} B_\eta S_{i,\alpha} L_{b;\eta\alpha\rho} \mathcal{O}_{j,\rho}^+ \quad (95)$$

projects to eq. (91) with

$$K_{b;\eta\alpha a}^{B,+} = \sum_{\rho} L_{b;\eta\alpha\rho} o_{j,\rho a}. \quad (96)$$

This is the even-sector analogue of the PJP step.

Gauge-covariant symmetry rule. The magnetic field is an axial vector, so it transforms with R_g , not Q_g . Therefore

$$K_{gb;\nu\gamma c}^{B,+} = \sum_{\eta, \alpha, a} R_{g,\eta\nu} R_{g,\alpha\gamma} K_{b;\eta\alpha a}^{B,+} \left[D_{g,+}^{(j)} \right]_{ca}. \quad (97)$$

Here $D_{g,+}$ is the adjoint action restricted to the time-even qutrit sector.

Inversion partner. Under inversion, B is axial and hence does not change sign. S is also axial. Thus the odd-bond sign is just the time-even qutrit mirror matrix:

$$\text{odd-bond sign matrix for } BS\lambda^+ : \quad M_+ = \text{diag}(+1, +1, -1, -1, +1). \quad (98)$$

For the off-diagonal subset $(\lambda^1, \lambda^4, \lambda^6)$,

$$M_{+, \text{off}} = \text{diag}(+1, -1, -1). \quad (99)$$

So

$$\lambda^1, \lambda^3, \lambda^8 : \text{same sign on the inversion partner}, \quad (100)$$

$$\lambda^4, \lambda^6 : \text{opposite sign on the inversion partner}. \quad (101)$$

Transported-gauge orbit formula, off-diagonal version. For reference column vectors $\mathbf{b}_{\eta,1}$, $\mathbf{b}_{\eta,4}$ and $\mathbf{b}_{\eta,6}$,

$$H_{\text{orbit,off}}^{BS\lambda}(t) = \sum_{i=0}^3 \sum_{\nu} B_{\nu}(t) \mathbf{S}_i^T R_i \sum_{\eta} R_{i,\eta\nu} \left[\mathbf{b}_{\eta,1}(\lambda_i^1 + \lambda_{3-i}^1) + \mathbf{b}_{\eta,4}(\lambda_i^4 - \lambda_{3-i}^4) + \mathbf{b}_{\eta,6}(\lambda_i^6 - \lambda_{3-i}^6) \right]. \quad (102)$$

If the diagonal even operators are included, add

$$\mathbf{b}_{\eta,3}(\lambda_i^3 + \lambda_{3-i}^3) + \mathbf{b}_{\eta,8}(\lambda_i^8 + \lambda_{3-i}^8). \quad (103)$$

to the bracket.

Implementation prescription

For $BS\lambda^+$, the inversion signs are not the same as the static K^- term. They are

$$\lambda^1 : +1, \quad \lambda^4 : -1, \quad \lambda^6 : -1 \quad (104)$$

for the off-diagonal sector. Proper-coset partners require Fe row signs R_i and magnetic-field axial-index signs R_i .

13.3 Fe–Fe–Tm term: $S + S + \lambda$

A symmetry-allowed term with two Fe spins and one Tm qutrit operator is

$$H_b^{SS\lambda} = \sum_{m,n} \sum_{\alpha,\beta,a \in +} S_{m,\alpha} S_{n,\beta} W_{b;mn;\alpha\beta a} \lambda_j^a. \quad (105)$$

Because two Fe spins are time-reversal even as a product, the Tm operator must again be time-reversal even:

$$a \in \{1, 3, 4, 6, 8\}. \quad (106)$$

For the onsite version used in many minimal simulations, $m = n = i$ and

$$H_{b,\text{onsite}}^{SS\lambda} = \sum_{a \in +} \mathbf{S}_i^T W_b^a \mathbf{S}_i \lambda_j^a. \quad (107)$$

For a single classical spin on the same site, only the symmetric part of W^a matters, so one may parameterize

$$W^a = \begin{pmatrix} W_{xx}^a & W_{xy}^a & W_{xz}^a \\ W_{xy}^a & W_{yy}^a & W_{yz}^a \\ W_{xz}^a & W_{yz}^a & W_{zz}^a \end{pmatrix}. \quad (108)$$

This is the origin of the six real parameters (xx, yy, zz, xy, xz, yz) per λ^a channel in a code implementation.

If the term is derived from a physical time-even Tm multipole $\mathcal{O}_\rho^+$, then

$$P_j \mathcal{O}_\rho^+ P_j = o_{j,\rho 0} \mathbf{1} + \sum_{a \in +} o_{j,\rho a} \lambda_j^a, \quad (109)$$

and the physical coupling

$$\sum_{\alpha,\beta,\rho} S_{m,\alpha} S_{n,\beta} W_{b;mn;\alpha\beta\rho} \mathcal{O}_{j,\rho}^+ \quad (110)$$

projects to eq. (105) with

$$W_{b;mn;\alpha\beta a} = \sum_{\rho} W_{b;mn;\alpha\beta\rho} o_{j,\rho a}. \quad (111)$$

Gauge-covariant symmetry rule. For the general two-Fe-site case,

$$W_{gb;\gamma\delta c} = \sum_{\alpha,\beta,a} R_{g,\alpha\gamma} R_{g,\beta\delta} W_{b;\alpha\beta a} \left[D_{g,+}^{(j)} \right]_{ca}. \quad (112)$$

For the onsite matrix form $\mathbf{S}^T W^a \mathbf{S}$, this becomes

$$W_{gb}^c = \sum_a \left[D_{g,+}^{(j)} \right]_{ca} R_g^T W_b^a R_g. \quad (113)$$

Since the R_i used here are diagonal π -rotation matrices, $R_i^T = R_i$.

Inversion partner. There is no polar electric prefactor and no odd physical J operator. Hence the inversion sign is again just the time-even qutrit parity matrix M_+ :

$$\boxed{\text{odd-bond sign matrix for } SS\lambda^+ : \quad M_+ = \text{diag}(+1, +1, -1, -1, +1).} \quad (114)$$

Thus $\lambda^1, \lambda^3, \lambda^8$ couple to inversion-even Tm combinations, whereas λ^4, λ^6 couple to inversion-odd Tm combinations.

Transported-gauge onsite orbit formula. Keeping only the off-diagonal even coherences for compactness,

$$\boxed{H_{\text{orbit,off}}^{SS\lambda} = \sum_{i=0}^3 \left[\mathbf{S}_i^T R_i W^1 R_i \mathbf{S}_i (\lambda_i^1 + \lambda_{3-i}^1) \right.} \\ \left. + \mathbf{S}_i^T R_i W^4 R_i \mathbf{S}_i (\lambda_i^4 - \lambda_{3-i}^4) \right.} \\ \left. + \mathbf{S}_i^T R_i W^6 R_i \mathbf{S}_i (\lambda_i^6 - \lambda_{3-i}^6) \right].} \quad (115)$$

If λ^3 and λ^8 are retained, add

$$\mathbf{S}_i^T R_i W^3 R_i \mathbf{S}_i (\lambda_i^3 + \lambda_{3-i}^3) + \mathbf{S}_i^T R_i W^8 R_i \mathbf{S}_i (\lambda_i^8 + \lambda_{3-i}^8). \quad (116)$$

13.4 Sign-summary table for implementation

For one transported-gauge inversion pair, the relative sign on the odd bond is the product of three contributions:

$$\boxed{\text{odd-bond sign} = (\text{external-field inversion parity}) \times (\text{local qutrit mirror parity}).} \quad (117)$$

Fe spin is axial, so it does not add an inversion sign. However, proper cosets still require R_i row signs for Fe components if Fe spins are stored in global axes.

Term	Tm sector	Odd-bond sign matrix	Surviving inversion-even Tm combination
$SJ \rightarrow S\lambda^-$	(2, 5, 7)	$M_- = (+, -, -)$	λ^2
$ESJ \rightarrow ES\lambda^-$	(2, 5, 7)	$-M_- = (-, +, +)$	λ^5, λ^7
$BS\lambda_{\text{off}}^+$	(1, 4, 6)	$M_{+, \text{off}} = (+, -, -)$	λ^1
$SS\lambda_{\text{off}}^+$	(1, 4, 6)	$M_{+, \text{off}} = (+, -, -)$	λ^1
$BS\lambda^+ \text{ or } SS\lambda^+$	(1, 3, 4, 6, 8)	$(+, +, -, -, +)$	$\lambda^1, \lambda^3, \lambda^8$

Warning

The phrase “surviving inversion-even Tm combination” refers only to the pair sum/ difference induced by the inversion partner. It does not say that the full response is nonzero for every Fe magnetic pattern or every field polarization. Proper-coset row signs, field-index signs, and the actual Fe mode pattern can impose additional cancellations.

13.5 Complete Hamiltonian with every term written out

Collecting the Fe sector, the onsite Tm sector, the leading bilinear K^- , the three higher-order field/two-spin vertices, and the external-field drives, the full transported-gauge model is

$$\boxed{H(t) = H_{\text{Fe}} + H_{\text{Tm},0} + \sum_{\nu} \left(H_{\nu}^{K^-} + H_{\nu}^{ESJ}(t) + H_{\nu}^{BS\lambda}(t) + H_{\nu}^{SS\lambda} \right) + H_{B,\text{Fe}}(t) + H_{B,\text{Tm}}(t) + H_{E,\text{Tm}}(t),} \quad (118)$$

where ν runs over inequivalent Fe-Tm bond orbits and $i = 0, 1, 2, 3$ over the four sublattices, with the proper-coset row matrices

$$R_0 = \text{diag}(+, +, +), \quad R_1 = \text{diag}(+, -, -), \quad R_2 = \text{diag}(-, +, -), \quad R_3 = \text{diag}(-, -, +). \quad (119)$$

Every term is now written explicitly.

Fe sector (first/second neighbour exchange and single-ion anisotropy).

$$\begin{aligned} H_{\text{Fe}} = & \sum_{\langle mn \rangle_1} [J_{1xy}(S_m^x S_n^x + S_m^y S_n^y) + J_{1z} S_m^z S_n^z] + \sum_{\langle mn \rangle_2} [J_{2xy}(S_m^x S_n^x + S_m^y S_n^y) + J_{2z} S_m^z S_n^z] \\ & + \sum_m [D_x (S_m^x)^2 + D_y (S_m^y)^2 + D_z (S_m^z)^2]. \end{aligned} \quad (120)$$

Onsite Tm sector (three-singlet qutrit).

$$H_{\text{Tm},0} = \sum_j \left[-\frac{\Delta_{12}}{2} \lambda_j^3 + \frac{\Delta_{12}-2\Delta_{13}}{2\sqrt{3}} \lambda_j^8 \right], \quad \Delta_{12} \approx 0.52 \text{ THz}, \quad \Delta_{13} \approx 1.13 \text{ THz}. \quad (121)$$

Static bilinear K^- (columns $\mathbf{k}_{\nu,2}, \mathbf{k}_{\nu,5}, \mathbf{k}_{\nu,7}$), λ^- sector.

$$H_{\nu}^{K^-} = \sum_{i=0}^3 \mathbf{S}_i^T R_i [\mathbf{k}_{\nu,2}(\lambda_i^2 + \lambda_{3-i}^2) + \mathbf{k}_{\nu,5}(\lambda_i^5 - \lambda_{3-i}^5) + \mathbf{k}_{\nu,7}(\lambda_i^7 - \lambda_{3-i}^7)]. \quad (122)$$

Electric-field-assisted ESJ (columns $\mathbf{e}_{\eta,a}$), λ^- sector.

$$H_{\nu}^{ESJ}(t) = \sum_{i=0}^3 \sum_{\eta} E_{\eta}(t) \mathbf{S}_i^T R_i [\mathbf{e}_{\eta,2}(\lambda_i^2 - \lambda_{3-i}^2) + \mathbf{e}_{\eta,5}(\lambda_i^5 + \lambda_{3-i}^5) + \mathbf{e}_{\eta,7}(\lambda_i^7 + \lambda_{3-i}^7)], \quad (123)$$

where the polar field carries $\mathbf{e}_{\eta,a} \rightarrow \sum_{\kappa} Q_{i,\eta\kappa} \mathbf{e}_{\kappa,a}$ with $Q_I = -1$ (this is the origin of the relative $-M_-$ versus K^-).

Magnetic-field-assisted $BS\lambda$ (columns $\mathbf{b}_{\eta,a}$), λ^+ sector.

$$\begin{aligned} H_{\nu}^{BS\lambda}(t) = & \sum_{i=0}^3 \sum_{\eta} B_{\eta}(t) \mathbf{S}_i^T R_i \left[\mathbf{b}_{\eta,1}(\lambda_i^1 + \lambda_{3-i}^1) + \mathbf{b}_{\eta,4}(\lambda_i^4 - \lambda_{3-i}^4) + \mathbf{b}_{\eta,6}(\lambda_i^6 - \lambda_{3-i}^6) \right. \\ & \left. + \mathbf{b}_{\eta,3}(\lambda_i^3 + \lambda_{3-i}^3) + \mathbf{b}_{\eta,8}(\lambda_i^8 + \lambda_{3-i}^8) \right], \end{aligned} \quad (124)$$

with $\mathbf{b}_{\eta,a} \rightarrow \sum_{\kappa} R_{i,\eta\kappa} \mathbf{b}_{\kappa,a}$ (axial field, hence R_i not Q_i).

Two-Fe-spin $SS\lambda$ (onsite symmetric matrices W^a), λ^+ sector.

$$\begin{aligned} H_{\nu}^{SS\lambda} = & \sum_{i=0}^3 \left[(\mathbf{S}_i^T R_i W^1 R_i \mathbf{S}_i)(\lambda_i^1 + \lambda_{3-i}^1) + (\mathbf{S}_i^T R_i W^4 R_i \mathbf{S}_i)(\lambda_i^4 - \lambda_{3-i}^4) + (\mathbf{S}_i^T R_i W^6 R_i \mathbf{S}_i)(\lambda_i^6 - \lambda_{3-i}^6) \right. \\ & \left. + (\mathbf{S}_i^T R_i W^3 R_i \mathbf{S}_i)(\lambda_i^3 + \lambda_{3-i}^3) + (\mathbf{S}_i^T R_i W^8 R_i \mathbf{S}_i)(\lambda_i^8 + \lambda_{3-i}^8) \right]. \end{aligned} \quad (125)$$

External-field drives.

$$H_{B,\text{Fe}}(t) = g_{\text{Fe}}\mu_B \sum_m \mathbf{B}(t) \cdot \mathbf{S}_m, \quad (126)$$

$$H_{B,\text{Tm}}(t) = -g_J\mu_B \sum_j \sum_{a \in \{2,5,7\}} \left[\sum_{\alpha,\beta} B_\alpha(t) (R_j)_{\alpha\beta} \mu_{0,\beta a} \right] \lambda_j^a, \quad (127)$$

$$H_{E,\text{Tm}}(t) = \sum_j \sum_{a \in \{1,3,4,6,8\}} \left[\sum_\eta E_\eta(t) p_{j,\eta a} \right] \lambda_j^a, \quad (128)$$

where μ_0 is the projected-moment matrix (19) and $p_{j,\eta a}$ is the projected Tm electric-multipole tensor (time-even, hence λ^+).

Time-reversal sector bookkeeping. The λ^- sector $\{2, 5, 7\}$ is fed by K^- and ESJ ; the λ^+ sector $\{1, 3, 4, 6, 8\}$ is fed by $BS\lambda$, $SS\lambda$, and $H_{E,\text{Tm}}$. $H_{\text{Fe}}, H_{\text{Tm},0}, H^{K^-}, H^{SS\lambda}, H_{B,\text{Fe}}, H_{B,\text{Tm}}$ are time-even by construction; the field-assisted terms are time-even only together with their explicit external field. This sorting is exactly what the cross-peak analysis of section 16 acts on.

14 Implementation remedy

14.1 If Fe spins are stored in global axes

Use the transported-gauge local λ_j basis for Tm. Then for each bond in a symmetry orbit:

$$\boxed{K_{\text{bond}}^- = R_{\text{Fe}(\text{bond})} K_\nu^- M_{\text{inv}},} \quad (129)$$

where

$$M_{\text{inv}} = \begin{cases} \mathbf{1}, & \text{direct/proper-coset branch,} \\ M_- = \text{diag}(+1, -1, -1), & \text{inversion-related branch.} \end{cases} \quad (130)$$

This is the version that matches a code convention in which Fe spins are lab Cartesian and Tm SU(3) states are local CEF-basis variables.

14.2 If Fe spins are instead stored in sublattice local frames

Then the R_i row signs may be absorbed into the spin variables themselves. In that convention the bond tensor should not multiply by R_i again. The code and the theory must choose one convention and use it consistently.

Warning

Do not mix these conventions. If Fe spins are global, include the R_i row signs in the bond tensor. If Fe spins are local, do not include them again. The Tm λ_j operators remain local qutrit coordinates either way.

14.3 Do not use $\mu^{-1}R\mu$ as a qutrit frame by default

The safe object is $P_j J_\alpha P_j$ or, in local coordinates, $\mu_{j,\alpha a} \lambda_j^a$. If a global observable is needed, compute it through

$$J_{\alpha,j}^{\text{global}} = \sum_{\beta,a} (R_j)_{\alpha\beta} \mu_{0,\beta a} \lambda_j^a \quad (131)$$

in transported gauge, or through the fully general W_g, D_g construction. Avoid treating

$$\mu^{-1}R\mu \quad (132)$$

as a genuine SU(3) adjoint representation unless it has been verified to be a proper orthogonal adjoint action induced by a unitary W .

15 Diagrammatic summary

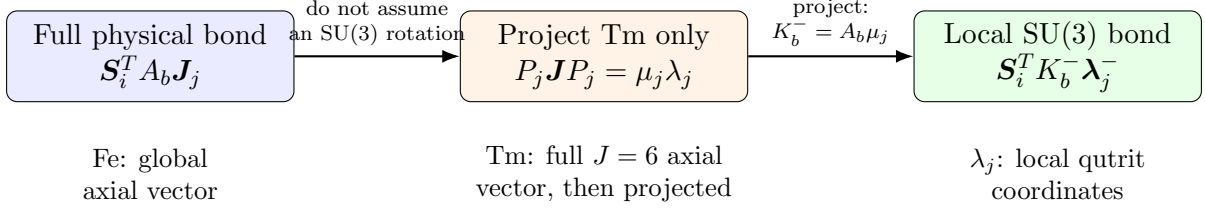


Figure 1: The correct order of operations. The Tm angular momentum is projected only after the physical symmetry constraints are formulated.

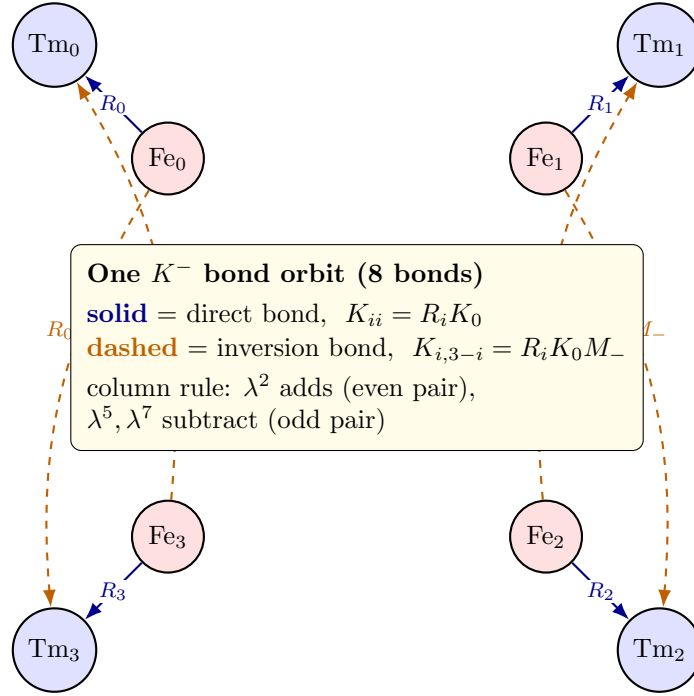


Figure 2: One transported-gauge K^- orbit. Fe sites (red) sit on inversion centres; Tm sites (blue) are exchanged in pairs by inversion ($\text{Tm}_0 \leftrightarrow \text{Tm}_3$, $\text{Tm}_1 \leftrightarrow \text{Tm}_2$). Each Fe participates in one *direct* bond (solid, coefficient $R_i K_0$) to its own Tm and one *inversion* bond (dashed, coefficient $R_i K_0 M_-$) to the partner Tm. Summing the orbit gives the even/odd column rule of (60).

16 Modeling the 2DCS cross peak: what must a vertex do?

The earlier sections fixed the *form* of every symmetry-allowed Fe–Tm coupling. We now ask the sharp dynamical question: which of them can actually produce the measured (q_{AFM}, E_{12}) 2DCS cross peak? The strategy is two-step. First we reduce the whole experiment to a *single* sharp requirement on an effective Tm force (this section). Then we test each candidate coupling against it as a Feynman-diagram catalogue (section 17). The experimental context is third-order magnetic-dipole THz 2DCS of orthoferrite magnons [1]; the language is standard nonlinear response theory [2].

16.1 The observable and M_{NL} , precisely

Two collinear THz pulses give the total drive $B(t) = B_A(t + \tau) + B_B(t)$. The detected magnetization is a functional of that field; expand it in powers of the drive,

$$M[B] = \sum_{n \geq 1} M^{(n)}[\underbrace{B, \dots, B}_n]. \quad (133)$$

With $M_A \equiv M[B_A]$, $M_B \equiv M[B_B]$, $M_{AB} \equiv M[B_A + B_B]$, the isolated nonlinear signal is

$$M_{\text{NL}}(t, \tau) = M_{AB} - M_A - M_B. \quad (134)$$

Order by order, $M^{(n)}[B_A + B_B] - M^{(n)}[B_A] - M^{(n)}[B_B]$ keeps only the *mixed* terms, those with at least one factor of each pulse:

$$n = 1: \quad 0 \quad (\text{linear, no mixed term}), \quad (135)$$

$$n = 2: \quad 2 M^{(2)}[B_A, B_B] \quad (\chi^{(2)}\text{-type, } \propto B_A B_B), \quad (136)$$

$$n = 3: \quad M^{(3)} \text{ mixed} \quad (\chi^{(3)}\text{-type, } B_A^2 B_B + B_A B_B^2). \quad (137)$$

So M_{NL} is purely nonlinear and starts at second order; the $n = 1$ piece cancels identically. For magnetic-dipole driving in a centrosymmetric magnet the leading robust contribution is third order [1]. The detected quantity at the target is the Tm c -axis dipole, $m_z^{\text{Tm}} \propto \lambda^2$, so the object of interest is $M_{\text{NL}}^{\lambda^2}(t, \tau)$, whose 2D transform should peak at $(\omega_\tau, \omega_t) = (q_{\text{AFM}}, E_{12})$.

16.2 The detection channel: only λ^2 is bright

In the E_{12} block the coherence is carried by $\lambda^1 = \text{Re}|1\rangle\langle 2|$ and $\lambda^2 = \text{Im}|1\rangle\langle 2|$, both oscillating at $\omega_{12} = E_{12}$, while λ^3 is the population imbalance. From (19) the radiated moment is

$$m_z^{\text{Tm}} \propto P_j J_z P_j = 5.264 \lambda^2, \quad (138)$$

so *only* λ^2 is magnetically bright. The onsite Hamiltonian $H_{\text{Tm},0} \supset -\frac{\Delta_{12}}{2} \lambda^3$ generates, through $[\lambda^3, \lambda^1] = 2i\lambda^2$, the planar precession

$$\dot{\lambda}^1 = -\Delta_{12} \lambda^2, \quad \dot{\lambda}^2 = +\Delta_{12} \lambda^1. \quad (139)$$

A force on *either* λ^1 or λ^2 therefore launches the same ring at ω_{12} . We may treat the Tm E_{12} sector as a single driven oscillator,

$$\lambda^2(t) = \int_0^t \sin[\omega_{12}(t - t')] h^1(t') dt', \quad (140)$$

driven by the effective transverse field $h^1(t)$ that a conversion vertex exerts on λ^1 .

16.3 Reframing to a sharp question

Equation (140) factorizes the 2D map. The *detection*-axis (t) structure is fixed entirely by the Tm pole at E_{12} : an impulsive step force gives a $1 - \cos \omega_{12} t$ ring, a delta kick gives a $\sin \omega_{12} t$ ring. All the *delay*-axis (τ) information—in particular whether the peak sits at $\omega_\tau = q_{\text{AFM}}$ —lives in the τ -dependence of $h^1(t, \tau)$. Hence the entire experiment collapses to one question:

The sharp question

A coupling produces the (q_{AFM}, E_{12}) cross peak *iff* the effective Tm force it generates,

$$M_{\text{NL}}^{\lambda^2}(\tau, t) \propto f(\omega_q \tau) [1 - \cos \omega_{12} t], \quad (141)$$

has f oscillating at $\omega_q = q_{\text{AFM}}$. Equivalently, the force $h^1(t, \tau)$ on the bright λ^1 channel must be (i) nonzero by symmetry, (ii) able to reach λ^1/λ^2 , and (iii) carry an ω_q modulation along the delay τ .

Everything below is the evaluation of $h^1(t, \tau)$ for each candidate vertex.

16.4 Feynman rules: fields, charges, and the vertex requirement

We treat the problem as a tree-level field theory. The propagating “fields” (internal/external lines) are

line	object	properties
Fe magnon	δS_α	pole at ω_q ; time-odd; the pump excites δS_y
Tm qutrit	λ^a	poles at CEF gaps; $\lambda^- = \{2, 5, 7\}$ (T-odd), $\lambda^+ = \{1, 3, 4, 6, 8\}$ (T-even)
B-photon	$B_\eta(t)$	external; time-odd, axial
E-photon	$E_\eta(t)$	external; time-even, polar

Every Fe magnon line is born from a pulse through the Zeeman source $g_{\text{Fe}}\mu_B \mathbf{B} \cdot \mathbf{S}$, so each external Fe leg costs one factor of the drive. Three “charges” are conserved at every vertex:

charge	rule at a vertex
energy ω	a static vertex conserves it $\Rightarrow$ connects only equal frequencies; an external leg injects $\hbar\omega_{\text{drive}}$
$\mathcal{T}$ -parity	each term is time-even: λ^- needs an <i>odd</i> number of T-odd legs (S, B); λ^+ an <i>even</i> number
irrep Γ	$\Gamma(\text{legs}) \otimes \Gamma(\lambda) \supset \Gamma_1$; the orbit sign sums of (60) are this projector

Detection and the M_{NL} subtraction add two more bookkeeping rules: the vertex must reach λ^1/λ^2 , and the diagram must attach at least one leg to each pulse. Collecting, a successful cross-peak vertex must pass all six tests:

Scorecard R1–R6

R1 (detection) reaches λ^1/λ^2 (the E_{12} block).

R2 (drive) is sourced by the pumped δS_y magnon.

R3 (parity) is $\mathcal{T}$ -even, which fixes the $\lambda^\pm$ sector.

R4 (irrep) is Γ_1 -allowed for the Γ_2/q_{AFM} geometry.

R5 (energy) can carry the $0.9 \rightarrow 0.5$ THz mismatch—needs ≥ 2 non- λ legs (two magnons, or one magnon + one photon).

R6 (M_{NL}) involves both pulses.

17 The vertex catalogue: a Feynman diagram for each coupling

We now write each candidate as an interaction vertex—a term in H with a coupling constant and a fixed set of legs—draw its tree diagram for the 2DCS process, and score it against R1–R6. In every diagram time runs left to right: a red solid line is an Fe magnon (ω_q), a blue dashed line a Tm coherence, a wavy arrow an external-field interaction (one pulse); a filled dot is a field-gated bilinear vertex, a filled square a two-spin vertex. The boxed expression above each diagram is the vertex itself.

17.1 K^- : the static bilinear (fails R3/R4/R5)

$$H_{K^-} = K_{a\alpha}^- S_\alpha \lambda^a, \quad a \in \{2, 5, 7\} \quad [1 \text{ Fe (odd)} + 1 \text{ Tm}^-(\text{odd}); \mathcal{T}\text{-even; no spare leg}] \quad (142)$$

This is a two-point Fe–Tm vertex; the only process it generates is a single Fe→Tm line conversion (fig. 3). It fails in three independent ways:

1. **(R5, linear).** K^- is linear in S , so it adds to the *quadratic* Hamiltonian—it is a basis rotation among the (harmonic) Fe and Tm normal modes. Coupled harmonic oscillators are diagonalizable and their 2D spectrum has *diagonal peaks only*. A single bilinear hop cannot transport a 0.9 THz coherence into a 0.5 THz one: a static vertex conserves energy, so it can only connect degenerate modes, i.e. it *hybridizes* rather than *converts*. The lone-pulse hop is in any case removed by the M_{NL} subtraction (R6).
2. **(R4, Γ_1).** Even the static torque vanishes: for the patterns F_x, G_z and the pumped δS_y the four-site orbit sums give $[0, 0, 0, 0]$ for every channel except $K_{2y}^- \rightarrow [8, 8, 8, 8]$ (section 18). All channels but $2y$ are Γ_1 -forbidden.
3. **(reorientation).** The one Γ_1 -allowed channel, K_{2y}^- , is a transverse molecular field on the order parameter; it reorients $\Gamma_2 \rightarrow G_y$ before any spectroscopic regime (lost between 0.085 and 0.090 meV). It is not a perturbative vertex.

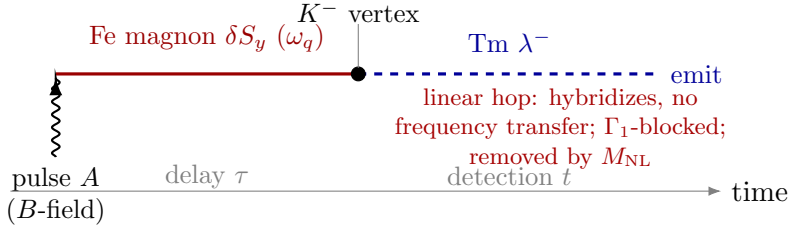


Figure 3: K^- (fails). A single Fe leg converts to a single λ^- leg—a linear hop. It only hybridizes the modes, the conversion matrix element is Γ_1 -cancelled, and the lone-pulse path is subtracted in M_{NL} .

The longitudinal temptation, and why it is still trapped

One might smuggle in two-magnon content through the *longitudinal* component $S_z = S_z^{(0)} - \delta S_\perp^2 / 2S_z^{(0)}$: a bilinear $K S_z \lambda$ does contain $\delta S_y^2 \lambda$. But S_z is time-odd, so $K S_z \lambda$ must place λ in λ^- —it is $K_{2z,5z,7z}^-$. The hidden δS_z carries the G_z sublattice sign pattern, so it cancels in the *same* orbit sum as the static G_z torque. The two-magnon signal is real but parity-trapped in λ^- and Γ_1 -cancelled—confirmed by the full nonlinear K_{2z}^- run (which contains the spin-length constraint) being dark to 10^{-15} even at 2.0 meV (section 18). To reach λ^1 the *same* δS_y^2 content must be carried by a $\mathcal{T}$ -even Fe operator: two spin legs (W) or a field leg (κ^B).

17.2 ESJ : electric-field-assisted (works, but wrong target—fails R1)

$$H_{ESJ}(t) = \kappa_{a\alpha\eta}^E E_\eta(t) S_\alpha \lambda^a, \quad a \in \{2, 5, 7\} \quad [1 \text{ E (even)} + 1 \text{ Fe (odd)} + 1 \text{ Tm}^-(\text{odd}); \mathcal{T}\text{-even}] \quad (143)$$

A genuine three-point vertex (fig. 4): pulse A’s magnetic field makes the Fe coherence, pulse B’s *electric* field enters at the vertex, and the E -photon supplies the energy mismatch (R5 ✓, R6 ✓, R2 ✓). But it feeds λ^- , and the surviving inversion-even combination under $-M_-$ is λ^5, λ^7 —the E_{13}

and E_{23} coherences, *not* E_{12} . So ESJ produces real cross peaks at (q_{AFM}, E_{13}) and (q_{AFM}, E_{23}) , but lands on the wrong readout for E_{12} (fails R1).

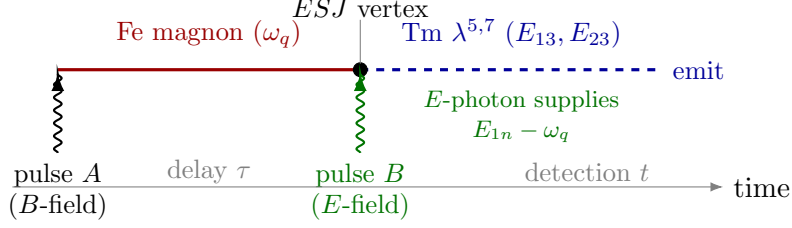


Figure 4: ESJ (works, wrong target). A clean three-point vertex; the E -photon carries the energy mismatch. But it feeds $\lambda^{5,7}$, so it lights up E_{13}/E_{23} , not the E_{12} target.

17.3 $W = SS\lambda$: the two-magnon vertex (passes all—the winner)

$$H_W = W_{\alpha\beta}^a S_\alpha S_\beta \lambda^a, \quad a \in \{1, 3, 4, 6, 8\} \quad [2 \text{ Fe (odd} \times \text{odd=even)} + 1 \text{ Tm}^+ \text{(even); } \mathcal{T}\text{-even; no photon}] \quad (144)$$

For the pump geometry the active term is $W_1^{yy} S_y^2 \lambda^1$ (R2: two y -magnon legs; R1: λ^1). This is a three-point vertex with *two* Fe legs, one sourced by each pulse (R6); see fig. 5. The vertex feels $h^1(t) = W_1^{yy} S_y(t)^2$ with $S_y(t) = A[\sin \omega_q(t + \tau) + \theta(t) \sin \omega_q t]$. The only term surviving M_{NL} is the two-pulse cross product $2S_y^a S_y^b$, which *rectifies*:

$$h_{\text{NL}}^1(t, \tau) = W_1^{yy} A^2 \left[\underbrace{\cos \omega_q \tau}_{\text{difference}} - \underbrace{\cos \omega_q (2t + \tau)}_{\text{sum}} \right] \theta(t). \quad (145)$$

The difference term is a t -independent step, born at the probe, which through (140) launches the $1 - \cos \omega_{12} t$ ring. Hence

$$M_{\text{NL}}^W(\tau, t) \propto W_1^{yy} A^2 \cos(\omega_q \tau) [1 - \cos \omega_{12} t]. \quad (146)$$

This is exactly the sharp-question form: it factorizes, oscillating at ω_q along τ and at ω_{12} along t (R3 with the delay modulation at ω_q), giving a single sharp peak at (q_{AFM}, E_{12}) with no diagonal partner (the $2\omega_q$ “sum” companion is off-resonant). The simulated λ^2 map has its rank-1 maximum exactly here, amplitude 51.0, SNR 117, the (E_{12}, E_{12}) diagonal $\sim 50\times$ weaker [7]; yy beats xx by $\sim 600\times$ (R2: S_x is the static F_x , barely fluctuating).

Where the E_{12} energy comes from (verified). Since δS_y^2 contains only 0 and $2\omega_q$, a static three-leg vertex between *free* modes could never emit at ω_{12} . The resolution is that the magnon legs are pulsed envelopes, not free modes: the rectified cross term is a *suddenly switched-on step*, and a step carries spectral weight at all frequencies in its edge, $\propto e^{-\omega_{12}^2 \sigma^2 / 2}$ for rise time σ . Quantum mechanically this is an intra-pulse difference-frequency (impulsive-Raman) process: the E_{12} quantum is drawn from the *pulse bandwidth*, while the magnon pair carries only the phase memory $\cos \omega_q \tau$. The mechanism is displacive and therefore requires nonadiabaticity—a falsifiable statement verified directly (section 19.5): stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades ($51.7 \rightarrow 1.7 \times 10^{-3}$), the adiabatic suppression the step picture demands, while switching the direct Tm drive off changes the peak by $< 10^{-4}$ relative (pure- W origin). The same sweep resolves the second Fe leg: for impulsive pulses it is the resonant probe magnon (peak $\propto A^2$); for longer pulses it crosses over to the *field-following quasistatic tilt* $\delta S^{\text{qs}}(t) \propto B(t)$ (peak $\propto A^1$)—both W -mediated, same peak location. The quasistatic leg also enables the *reverse* conversion $\lambda^{1,2}(\tau) \times \delta S^{\text{qs}}(\text{pulse } B) \rightarrow q_{\text{AFM}}$, an (E_{12}, q_{AFM}) feature in the Fe channel whenever a first-order E_{12} delay coherence exists (section 19).

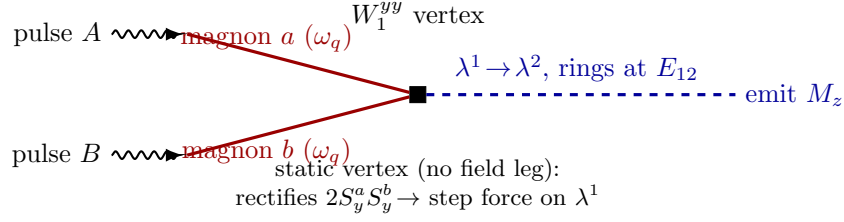


Figure 5: W_1^{yy} (winner). Each pulse makes one δS_y magnon; the two converge on the time-independent $S_y^2 \lambda^1$ vertex (square), whose two-pulse cross term rectifies to a step force that launches the $\lambda^1 \rightarrow \lambda^2$ ring at E_{12} . The response factorizes, $\cos(\omega_q \tau)[1 - \cos \omega_{12} t]$, giving the clean rank-1 cross peak.

17.4 $\kappa^B = BS\lambda$: magnetic-field-assisted (passes, but noisy)

$$H_{\kappa B}(t) = \kappa_{a\alpha\eta}^B B_\eta(t) S_\alpha \lambda^a, \quad a \in \{1, 3, 4, 6, 8\} \quad [1 \text{ B (odd)} + 1 \text{ Fe (odd)} + 1 \text{ Tm}^+(\text{even}); \mathcal{T}\text{-even}] \quad (147)$$

The active term is $\kappa_{1y}^B B_y S_y \lambda^1$. A three-point vertex with one B -photon, one Fe leg and one λ^+ leg (fig. 6); the B -photon supplies the mismatch (R5). It is field-gated: $h^1(t) = \kappa_{1y}^B \mathcal{B}_y(t) S_y(t)$ with $\mathcal{B}_y(t)$ sharply localized at the pulses. At the probe $t = 0$, $S_y(0) = A \sin \omega_q \tau$, so the kick $\propto \sin \omega_q \tau$ launches the same ring,

$$M_{\text{NL}}^{\kappa B}(\tau, t) \propto \kappa_{1y}^B \sin(\omega_q \tau) [1 - \cos \omega_{12} t], \quad (148)$$

again on target (R1–R6 ✓). This is the $BS\lambda^+$ analogue of ESJ , feeding λ^1 instead of $\lambda^{5,7}$. The price: the same gated vertex also dumps each pulse’s *local* Fe response into Tm at both pulse times, generating a strong (E_{12}, E_{12}) diagonal and a $(0, E_{12})$ background that dominate the bare cross peak (the on-target maximum is genuine but only rank-3 at zero carrier). A carrier resonant with a Tm transition cleans this up—a sweep amplifies the target $\sim 13\times$ at $\Omega_B = 0.70 \text{ THz}$ (E_{23}) [7]. The definitive resolution, however, is not a carrier but the *polarization* of the gate: the physical vertex is gated by the lab-frame field component $B_\eta(t)$, and its dominant symmetry slice is z -gated, so in the $H \parallel a$ geometry ($B_z = 0$) κ^B is identically silent and its noise never appears. Its proper home is the $H \parallel c$ geometry, where it produces the measured (E_{12}, q_{AFM}) cross peak; see section 19.

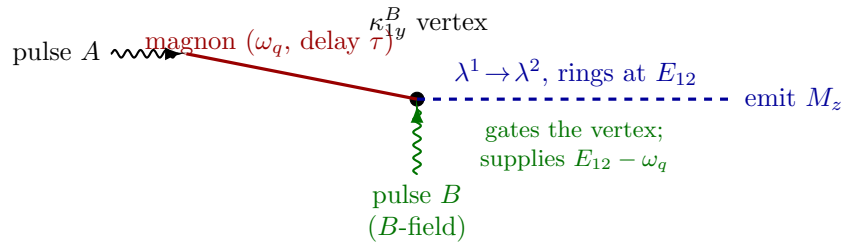


Figure 6: κ_{1y}^B (works, noisy). One magnon from pulse A meets the magnetic field of pulse B at the $S_y \lambda^1$ vertex; the B -photon gates the conversion and carries the mismatch. Same $\lambda^1 \rightarrow \lambda^2$ readout as W_1^{yy} , but it also feeds the (E_{12}, E_{12}) diagonal and a $(0, E_{12})$ background, so it needs a resonant carrier to be a robust target.

17.5 Scorecard

vertex	term	legs	sector	R1	R2	R3/4/5	verdict
K^-	$S\lambda^-$	$1\text{Fe}+1\text{Tm}^-$	λ^-	$\times$	—	$\times$	hybridizes / Γ_1 -blocked / reorients
ESJ	$ES\lambda^-$	$E+\text{Fe}+\text{Tm}^-$	λ^-	$\times$	$\checkmark$	$\checkmark$	real, but on E_{13}/E_{23}
W	$SS\lambda^+$	$2\text{Fe}+\text{Tm}^+$	λ^+	$\checkmark$	$\checkmark(yy)$	$\checkmark$	winner: rank-1, factorized
κ^B	$BS\lambda^+$	$B+\text{Fe}+\text{Tm}^+$	λ^+	$\checkmark$	$\checkmark(1y)$	$\checkmark$	works; B_z -gated, geometry 2

The two passes share the same readout, $\lambda^1 \rightarrow \lambda^2$ at E_{12} , and differ only in how the Fe motion is fed in: W uses *two* magnon legs (difference-frequency rectification), κ^B uses *one* magnon leg plus a photon. Both are the minimal $\mathcal{T}$ -even ways to deliver the δS_y^2 two-magnon signal onto the bright λ^1 —the step that K^- , trapped in λ^- , cannot take.

18 Numerical confirmation

The catalogue is confirmed by direct simulation in the calibrated $1 \times 1 \times 1$ mixed TmFeO_3 model. All 2DCS runs use Fe pump $H \parallel x$, $\tau \in [-40, 0]$, $t \in [-40, 40]$, and step $\Delta t = 0.02$ (code units); the Tm sector is not directly pumped unless stated.

18.1 Static one-at-a-time K^- scan (0.2 meV): the Γ_1 block

A static ground-state scan with one K^- component on at a time confirms R4 directly:

component	E/N (meV)	Fe order	$(\lambda_2, \lambda_5, \lambda_7)_{\text{uniform}}$	verdict
baseline	-42.496758	$F_x + G_z$	(0, 0, 0)	reference
K_{2x}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{2y}^-	-43.654818	G_y	(0.888320, 0, 0)	spin reorients
K_{2z}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{5x}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{5y}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{5z}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{7x}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{7y}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert
K_{7z}^-	-42.496758	$F_x + G_z$	(0, 0, 0)	inert

Only K_{2y}^- does anything, and it does so by leaving Γ_2 : an adiabatic ramp keeps Γ_2 stable through 0.085 meV and loses it by 0.090 meV, flipping to G_y with a large uniform λ_2 . The pattern is the exact orbit sign algebra (R_i with $M_- = \text{diag}(+1, -1, -1)$):

$$F_x : K_{2x}^-, K_{5x}^-, K_{7x}^- \rightarrow [0, 0, 0, 0], \quad G_z : K_{2z}^-, K_{5z}^-, K_{7z}^- \rightarrow [0, 0, 0, 0], \quad (149)$$

$$\delta S_y : K_{2y}^- \rightarrow [8, 8, 8, 8], \quad \delta S_y : K_{5y}^-, K_{7y}^- \rightarrow [0, 0, 0, 0]. \quad (150)$$

18.2 Fe-pump 2DCS on the inert channels: dark to machine precision

Running 2DCS on the inert channels (Fe pump only) and extracting the maximum nonlinear $\text{SU}(3)$ amplitude over the (t, τ) grid confirms R5—these are dark even with the *full* nonlinear dynamics (hence including the longitudinal δS_z rectification):

run	$\max M_{\text{NL}}^{\lambda_2} $	$\max M_{\text{NL}}^{\lambda_5} $	$\max M_{\text{NL}}^{\lambda_7} $
$K_{2x}^- = 0.2$	6.06×10^{-18}	0	0
$K_{2x}^- = 1.5$	2.96×10^{-16}	0	0
$K_{2z}^- = 0.2$	9.87×10^{-16}	0	0
$K_{2z}^- = 2.0$	6.86×10^{-15}	0	0
$K_{5y}^- = 0.2$	0	0	0
$K_{7y}^- = 0.2$	0	0	0

Up to $K_{2x}^- = 1.5 \text{ meV}$ and $K_{2z}^- = 2.0 \text{ meV}$ the induced λ_2 is at machine precision—the bilinear cannot, even nonlinearly, manufacture the E_{12} coherence. Adding a direct Tm $H \parallel x$ pump on top reproduces the ordinary Tm response and reveals no hidden K^- channel. By contrast, the no-damping scan over the higher-order set isolates exactly W_1^{yy} (rank-1, $|M_{\text{NL}}^{\lambda_2}| = 51.0$, SNR 117) and κ_{1y}^B as the on-target vertices [7], precisely the two that pass R1–R6.

19 The cross-polarization scenario: two geometries, one Hamiltonian

This section states the final, validated picture and derives the physics of every measured peak. We restrict to the *cross-polarization* readout of the two field geometries, because that is where the experiment isolates clean, well-separated cross peaks; the same-polarization channel of geometry A is analyzed separately in section 20, where it is shown to require one additional selection rule (a z -dark $1 \rightarrow 3$ CEF dipole) and nothing else.

The model is the calibrated Fe + Tm-qttrit Hamiltonian of section 13.5 driven by the *measured* single-cycle THz waveform (spectrum peaked at 0.53 THz, no DC component). A single set of three Fe–Tm couplings is switched on *simultaneously* and used, unchanged, for both geometries; nothing is retuned between $H \parallel a$ and $H \parallel c$, and the geometry alone selects which peaks appear. Each coupling carries one measured peak, and the isolated-vertex tests of sections 19.2 and 19.3 show none is redundant:

coupling	type	measured feature it produces
W_1^{yy} (two-magnon exchange)	static	G-A (q_{AFM}, E_{12}) cross peak
κ_{1y}^B (field-gated)	impulsive	G-B (E_{12}, q_{AFM}) transfer peak
W_1^{xz} (static hybridization)	static	G-B ($q_{\text{FM}}, q_{\text{AFM}}$) peak + post-drive buildup

All three are held at the same strength (0.01 in code units) in every run of the final scenario.

19.1 Geometries, excitation, and the experimental readout

The two measured configurations are

- **Geometry A**, $H \parallel a$, $E \parallel c$. The electric field $E \parallel c$ drives the Tm crystal-field (CEF) E_{12} transition through its c -axis electric dipole; the magnetic field $H \parallel a$ Zeeman-drives the antiferromagnetic magnon q_{AFM} .
- **Geometry B**, $H \parallel c$, $E \parallel a$. The magnetic field $H \parallel c$ drives the ferromagnetic-canting magnon q_{FM} ; the electric field $E \parallel a$ drives the Tm E_{12} transition through the orthogonal (a -axis) dipole component.

The measured pulse. The pump is the measured single-cycle THz waveform of fig. 7: a bipolar $\sim 1.2 \text{ ps}$ transient with no DC component, whose power spectrum peaks at 0.53 THz and covers every mode of interest. Ingested as a table (times converted to code units, $t_{\text{code}} = t_{\text{ps}}/0.6582$), it drives E_{12} at full weight and the magnons and higher CEF lines at the spectral amplitudes listed in table 2; the absence of a DC component removes the quasi-static tilt that would otherwise contaminate the transfer channels.

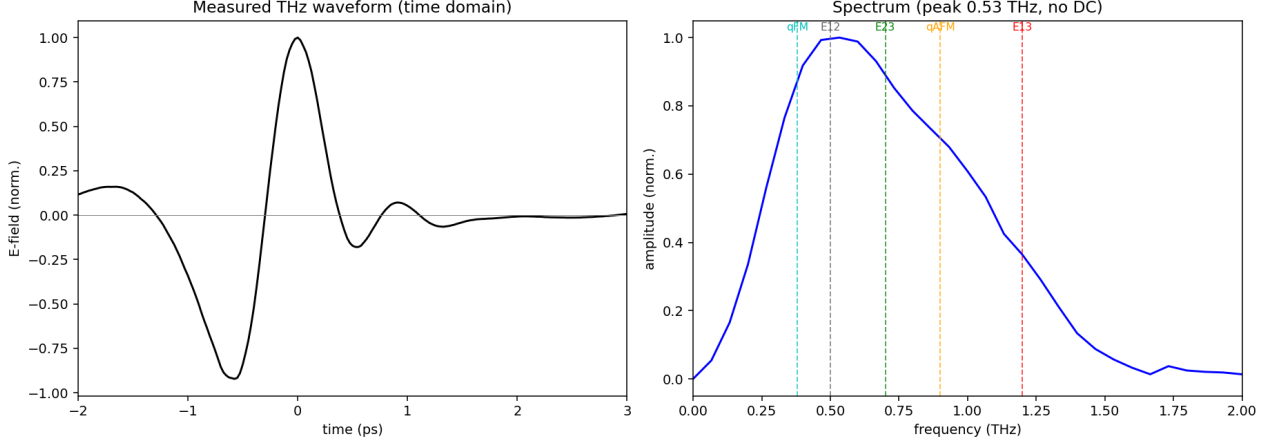


Figure 7: The measured THz pump. Left: time-domain waveform (single bipolar cycle, ~ 1.2 ps, no DC). Right: amplitude spectrum, peaked at 0.53 THz, with the five mode frequencies marked; the pulse covers E_{12} (amplitude 1.00) down through E_{13} (0.36).

The experimental readout. The detected signal is the emitted THz field, and the experiment supplies the absolute radiative weights. Every lineshape and emission parameter in this section is taken from the measurement; table 2 collects them together with the derived simulation inputs—the Bloch/Gilbert damping rates (a linewidth *is* a decay rate) and the $\sqrt{f}$ emission weights.

Table 2: Experimental inputs and the derived simulation parameters. CEF damping is Bloch on the Gell-Mann pair for each transition, $\gamma_a[\text{meV}] = \gamma[\text{THz}] \times h$; magnon damping is Gilbert, $\alpha = \gamma/2\nu$, fixed by the q_{AFM} line. Emission weight is the electric dipole $\sqrt{f}$ (E1) for the CEF lines and M1 for the magnons (whose E1 activity is negligible).

mode	ν [THz]	f	$\sqrt{f}$	γ [THz]	damping (sim)	pulse amp.
q_{FM}	0.38	0.08	0.28	0.005	$\alpha = 1.7 \times 10^{-3}$	0.92
q_{AFM}	0.90	7×10^{-4}	0.026	0.003	$\alpha = 1.7 \times 10^{-3}$	0.68
E_{12}	0.50	6.5	2.55	0.038	$\gamma(\lambda^{1,2}) = 0.157 \text{ meV}$	1.00
E_{23}	0.70	6.6	2.57	0.10	$\gamma(\lambda^{6,7}) = 0.414 \text{ meV}$	0.85
E_{13}	1.20	8.7	2.95	0.05	$\gamma(\lambda^{4,5}) = 0.207 \text{ meV}$	0.36

Two facts organize everything. First, the excitation scales as E^2 and the detection linearly in E : the signal is third order, with two excitation interactions and one emission. Second, the CEF transitions are electric-dipole (E1) bright ($\sqrt{f} \approx 2.5\text{--}3.0$) while the q_{AFM} magnon is E1-*dark* ($\sqrt{f} = 0.026$) but magnetic-dipole (M1) bright—it is a large-amplitude spin precession. Consequently a magnon can be *read out* in two distinct ways: (i) it borrows a bright CEF electric dipole by modulating the Tm coordinate through the exchange vertex, emitting at a CEF frequency (geometry A), or (ii) it radiates directly through its own magnetic dipole at the magnon frequency (geometry B). This distinction, not any retuning, is what separates the two geometries’ peak lists.

Throughout, the delay axis is plotted in the physical convention $\omega_\tau \rightarrow -\omega_\tau$; a peak on the $+\omega_\tau$ side is non-rephasing (NR), on the $-\omega_\tau$ side rephasing (R).

19.2 Geometry A cross peak: (q_{AFM} , E_{12})

Mechanism. During the delay τ the Zeeman-launched q_{AFM} magnon precesses at $\omega_\tau = 0.90$ THz. The static two-magnon exchange vertex W_1^{yy} —the term by which a magnon distortion modulates the Tm crystal-field splitting (section 17)—transcribes this magnon phase onto the bright Tm λ^2 (E_{12}) coordinate. During detection t the system therefore radiates at $\omega_t = E_{12} = 0.50$ THz through the strong Tm electric dipole. The peak is a *magnetolectric* cross peak: a magnon that is itself nearly invisible to the light field is detected by borrowing the Tm E1 dipole. This is exactly why

geometry A shows a bright peak whose emission sits at the CEF frequency rather than at the magnon frequency.

Result. At $T = 0$ the λ^2 readout gives, in units of the channel maximum,

$$(+q_{\text{AFM}}, E_{12}) = 1.00, \quad (-q_{\text{AFM}}, E_{12}) = 0.33, \quad \text{residuals} \leq 0.18.$$

The peak is dominant and non-rephasing, with a rephasing partner at 0.33: the symmetric W -vertex admits both time orderings of the magnon coherence, so the signal appears on both sides of ω_τ with the non-rephasing branch stronger. The one residual, a weak $(q_{\text{AFM}}, q_{\text{AFM}})$ shoulder at 0.18, also emits at the magnon frequency. Note that the experimental linewidths make this peak clean automatically: the E_{12} coherence dephases in ~ 8 ps ($\gamma_{E_{12}} = 0.038$ THz) whereas the q_{AFM} magnon lives ~ 100 ps ($\gamma_{q_{\text{AFM}}} = 0.003$ THz), so the magnon survives the delay and dominates while the fast-dephasing E_{12} diagonal is suppressed—no analysis-window truncation is needed.

19.3 Geometry B cross peaks: $(q_{\text{FM}}, q_{\text{AFM}})$ and (E_{12}, q_{AFM})

Here the readout is the magnon magnetic dipole (M1): both peaks emit at $\omega_t = q_{\text{AFM}} = 0.90$ THz.

$(q_{\text{FM}}, q_{\text{AFM}})$ — **a magnon–magnon cross peak.** $H \parallel c$ launches the q_{FM} magnon, which precesses at $\omega_\tau = 0.38$ THz during the delay. The ferro- and antiferromagnetic modes are coupled by the single-ion anisotropy and the Dzyaloshinskii–Moriya interaction, so the q_{FM} coherence is re-emitted at the q_{AFM} frequency during t . This is the purely magnetic cross peak, present whenever both magnon branches are dynamically coupled.

(E_{12}, q_{AFM}) — **a Tm→magnon transfer peak, made by the field-gated vertex.** $E \parallel a$ launches the Tm E_{12} coherence, precessing at $\omega_\tau = 0.50$ THz; a Fe–Tm vertex transfers it to the q_{AFM} magnon, which radiates via M1. Which vertex? Isolating the couplings settles it directly: with the field-gated κ_{1y}^B alone the cross peak is present, $(E_{12}, q_{\text{AFM}}) = 0.46$, whereas with the static W_1^{xz} alone it is *absent* (only the magnon–magnon peak survives). The E_{12} phase is imprinted on the magnon emission by the impulsive, field-gated transfer of κ_{1y}^B , which acts while $B_z(t) \neq 0$; the static hybridization does not generate this correlation. Because the transfer runs off the E_{12} excitation rather than the direct Zeeman drive, the peak is non-rephasing.

But the peak is not the whole story: the post-drive buildup requires a static vertex.

The experiment reports that the geometry-B signal keeps building up *after* the pump has passed—and a field-gated vertex cannot do this. Isolating the magnon envelope (fig. 8) makes the division of labor explicit: κ_{1y}^B alone leaves the post-pulse envelope flat (modulation 0.03), an impulsive kick that ends with the field, while the static W_1^{xz} keeps exchanging the persisting q_{FM} (and E_{12}) reservoir with the q_{AFM} magnon, so the envelope modulates by 0.33 and rides $\sim 1.25\times$ above the impulsive level long after the pulse. Neither vertex suffices alone: κ_{1y}^B creates the measured (E_{12}, q_{AFM}) cross peak but no buildup, W_1^{xz} creates the buildup (and the magnon–magnon peak) but not the transfer cross peak. Both are therefore kept, and this is why a W_1^{xz} -only model—though it would automatically give the observed buildup—is excluded: it loses the (E_{12}, q_{AFM}) peak entirely.

Result. At $T = 0$,

$$(+q_{\text{FM}}, q_{\text{AFM}}) = 1.00, \quad (+E_{12}, q_{\text{AFM}}) = 0.43, \quad \text{residuals} \leq 0.12.$$

Both designated peaks are non-rephasing and clean; nothing else exceeds 0.11. Note the Tm CEF channels λ^{4-7} are *identically* zero in geometry B—the $E \parallel a$, $H \parallel c$ drive closes on the $\{\lambda^1, \lambda^2, \lambda^3\}$ subalgebra—so no CEF emission competes in this channel.

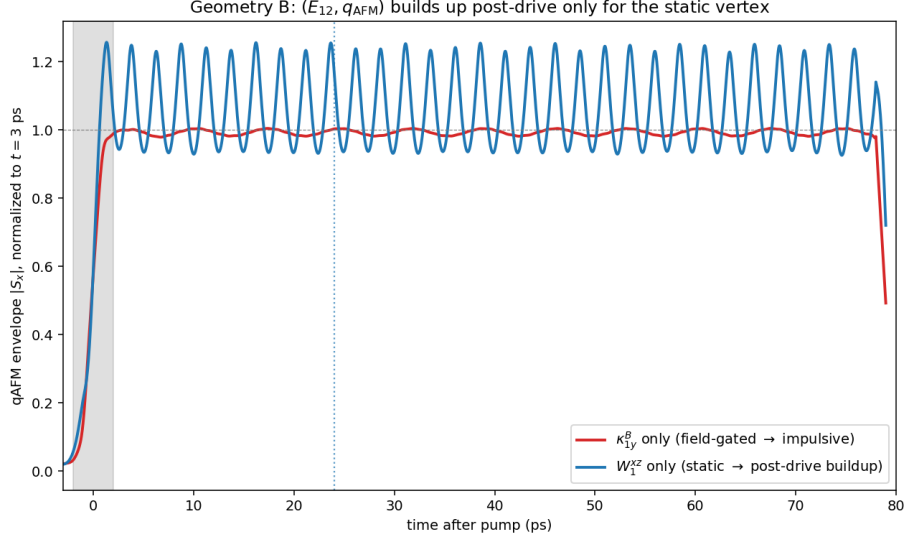


Figure 8: Why both a field-gated and a static vertex are needed. The q_{AFM} magnon envelope in geometry B, normalized to its value just after the pump. The field-gated κ_{1y}^B (red)—which alone produces the (E_{12}, q_{AFM}) cross peak—delivers an impulsive kick with a flat post-pulse envelope (modulation 0.03). The static W_1^{xz} (blue)—which alone cannot produce that cross peak—keeps exchanging the q_{FM}/E_{12} reservoir with the magnon, so its envelope modulates by 0.33 and rides $\sim 25\%$ above the impulsive level. The measured post-drive buildup demands the static vertex; the measured transfer peak demands the field-gated one.

19.4 Finite temperature

The finite-temperature response is the exact Boltzmann ensemble over the qutrit level seeds—a mixture of trajectories launched from $|1\rangle$, $|2\rangle$, $|3\rangle$ with weights $e^{-E_i/k_B T}$ —which for the linear single-ion drive is the density-matrix evolution. With $E_{12} = 2.07 \text{ meV} = 24 \text{ K}$ the first excited level begins to populate in the 5–10 K range, and the two geometries respond differently.

Geometry A. (q_{AFM}, E_{12}) remains the dominant peak through 10 K and the channel stays clean: the rephasing partner grows only mildly ($0.33 \rightarrow 0.39$) and the $(q_{\text{AFM}}, q_{\text{AFM}})$ shoulder from $0.18 \rightarrow 0.22$. Because both the excitation and the detection run through the long-lived magnon, the geometry-A response is nearly temperature-independent below $\sim 10 \text{ K}$.

Geometry B. The magnon–magnon peak $(q_{\text{FM}}, q_{\text{AFM}})$ is temperature-robust: it is a pure two-magnon process, independent of the qutrit populations. The transfer peak (E_{12}, q_{AFM}) , by contrast, is proportional to the ground-state population difference $n_1 - n_2$ that sets the E_{12} coherence amplitude, and it weakens monotonically: $0.43 (T=0) \rightarrow 0.41 (5 \text{ K}) \rightarrow 0.35 (8 \text{ K}) \rightarrow 0.29 (10 \text{ K})$. Both designated peaks remain clearly resolved through $\sim 8 \text{ K}$.

The working window. A measurement in the 5–8 K range keeps all three cross peaks— (q_{AFM}, E_{12}) in geometry A and both geometry-B peaks—dominant and well separated, while providing a falsifiable thermal signature: geometry A is temperature-flat while geometry B loses (E_{12}, q_{AFM}) relative to $(q_{\text{FM}}, q_{\text{AFM}})$ as T rises, tracking $n_1 - n_2$.

19.5 Everything from experiment: linewidths as damping, oscillator strengths as emission

The final picture (fig. 9, folder `tmfeo3_2dcs_final/EXPERIMENTAL_FINAL/`) is computed with no free lineshape parameters: every linewidth and every emission weight is taken directly from the measurement.

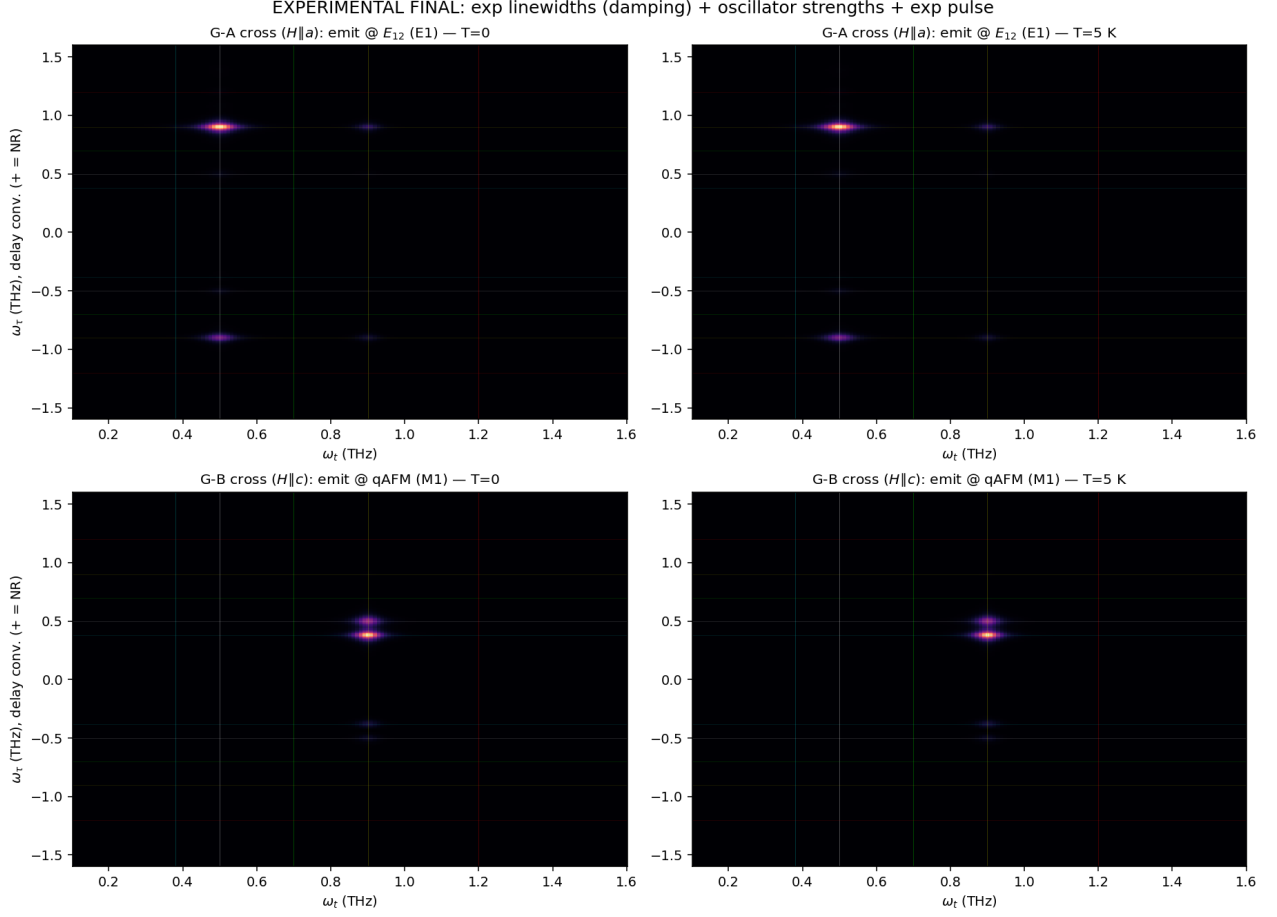


Figure 9: The cross-polarization scenario computed entirely from experimental inputs: the measured THz pulse, the measured linewidths used as the mode damping rates, and the measured oscillator strengths used as the emission weights. Top: geometry A ($H \parallel a$), λ^2 channel emitting at $E_{12} = 0.50$ THz through the bright Tm electric dipole; the dominant (q_{AFM}, E_{12}) peak with its weaker rephasing partner at -0.9 . Bottom: geometry B ($H \parallel c$), magnon S_x channel emitting at $q_{\text{AFM}} = 0.90$ THz through the M1 dipole; the magnon–magnon ($q_{\text{FM}}, q_{\text{AFM}}$) peak at $\omega_\tau = +0.38$ and the Tm→magnon (E_{12}, q_{AFM}) peak at $\omega_\tau = +0.50$. Left column $T = 0$, right column $T = 5$ K. Delay convention $+\omega_\tau =$ non-rephasing; the quasi-elastic band $|\omega_\tau| < 0.1$ THz is masked. Runs in `tmfeo3_2dcs_final/EXPERIMENTAL_FINAL/`.

Linewidths $\rightarrow$ damping. A measured linewidth *is* a decay rate, so each mode is damped at its experimental value rather than by an ad hoc constant. The CEF coherences receive Bloch damping $\gamma_a = (\text{linewidth in THz}) \times h$ on the corresponding Gell-Mann components— $\gamma(\lambda^{1,2}) = 0.157$ meV for E_{12} (0.038 THz), $\gamma(\lambda^{4,5}) = 0.207$ meV for E_{13} (0.05 THz), $\gamma(\lambda^{6,7}) = 0.414$ meV for E_{23} (0.10 THz). The magnons receive Gilbert damping $\alpha = \Delta\nu/2\nu$, fixed by the antiferromagnetic linewidth to $\alpha = 0.003/(2 \times 0.9) = 1.67 \times 10^{-3}$. This single change removes the need for the analysis-window truncation used earlier: the disparate lifetimes—magnon ~ 100 ps, $E_{12} \sim 8$ ps—are now generated by the dynamics, and the sharp q_{AFM} line and the fast-dephasing CEF diagonals emerge with the correct relative weights.

Oscillator strengths $\rightarrow$ emission. The detected field weights each mode by its electric dipole $\sqrt{f}$ for the CEF lines ($\sqrt{f_{E_{12}}} = 2.55$, etc.) and by the magnetic dipole (M1) for the magnons; the magnon electric activity $f_{q_{\text{AFM}}} = 7 \times 10^{-4}$ is negligible, so q_{AFM} never radiates electrically—it is seen in geometry A only by lending its phase to the Tm E_{12} dipole, and in geometry B only through its own M1 dipole.

The remaining numerics are as before: $1 \times 1 \times 1$ mixed lattice, RK4 ($\Delta t = 0.02$), delay $\tau \in [-120, 0]$, detection to $t_{\text{max}} = 160$, honest M_1 subtraction, no pulse-window chunking, the measured

pump waveform, the two-magnon vertex W_1^{yy} , the static hybridization W_1^{xz} (source of the post-drive buildup, fig. 8), and the geometry-B gate κ_{1y}^B . Spectra use a Hann apodization on both axes to suppress the sidelobes of the now long-lived magnon coherence.

19.6 Falsifiable predictions

1. **Emission-frequency fingerprint.** The geometry-A cross peak emits at $E_{12} = 0.50$ THz (a CEF line), whereas the geometry-B peaks emit at $q_{\text{AFM}} = 0.90$ THz (the magnon line). A magnon read out through a CEF dipole versus through its own M1 dipole is the structural signature of the two mechanisms.
2. **Linewidth discriminator.** Any peak emitting at q_{AFM} inherits the magnon linewidth (0.003 THz), a factor ~ 15 sharper than a CEF emission (E_{12} : 0.038 THz). The geometry-B peaks should be spectrally sharp along ω_t ; the geometry-A peak broad.
3. **Thermal trend.** $(q_{\text{FM}}, q_{\text{AFM}})$ is temperature-flat; (E_{12}, q_{AFM}) falls as $n_1 - n_2$; (q_{AFM}, E_{12}) acquires a q_{FM} -assisted partner above ~ 8 K. A temperature scan separates the magnon–magnon from the transfer peaks.
4. **One Hamiltonian.** The same couplings produce the E_{12} -emitting peak in geometry A and the q_{AFM} -emitting peaks in geometry B with no retuning; the geometry (field orientation) alone selects the readout channel.

20 The same-polarization channel of geometry A: five peaks from a z -dark $1 \rightarrow 3$ dipole

The same-polarization detection channel of geometry A ($H \parallel a$, $E \parallel c$ in, $E \parallel c$ out) shows, in (ω_τ, ω_t) and THz, its largest signals at

$$(0.5, 0.7) \approx (0.5, 0) > (0.5, 1.2) > (0.9, 0.38), (0.9, 1.2),$$

with *both* members of the 0.5-family transfer pair on the *same* signed side of the delay axis. This section derives the physics of each peak and reports the final calibrated simulation (run `it14hr`, thermal Boltzmann ensemble over the CEF levels, measured pulse, measured linewidth dampings). Throughout, peaks are identified *blindly*: a feature counts only if it is a genuine two-dimensional local maximum with prominence ≥ 1.5 over the local median background; no box-probing of preassigned positions is used.

20.1 The structural clue: no $\omega_\tau = 1.2$ rows anywhere

The experimental map contains delay frequencies $\omega_\tau \in \{0.5, 0.9\}$ only. There is no peak, strong or weak, with delay phase $E_{13} = 1.2$ THz. Since a first-order action of the pump on a bright $1 \rightarrow 3$ dipole would necessarily store an E_{13} coherence during the delay—and the same matrix element would produce a $(E_{13}, E_{23}) = (1.2, 0.7)$ transfer peak with *exactly* the same dipole product $\mu_{12}\mu_{13}\mu_{23}$ as the observed $(E_{12}, E_{23}) = (0.5, 0.7)$ peak—the absence of the entire $\omega_\tau = 1.2$ row is a selection rule, not an accident:

$$\mu_{13}^z \simeq 0 : \quad \text{the } 1 \rightarrow 3 \text{ CEF transition is electric-dipole dark for } E \parallel c. \quad (151)$$

This is the natural consequence of a parity-alternating CEF ladder under the relevant mirror: with $|1\rangle$ even, $|2\rangle$ odd, $|3\rangle$ even, the (m_z -odd, exchange-induced) z -dipole connects $1 \leftrightarrow 2$ and $2 \leftrightarrow 3$ but *not* $1 \leftrightarrow 3$. It does not contradict the measured 1D oscillator strength of E_{13} ($f = 8.7$, table 2): that strength lives in the orthogonal (a/b -axis) dipole components—precisely the ones the cross-polarized channel of section 19 uses. The same transition is bright in one polarization and dark in the other.

In the simulation the rule is implemented by removing the λ^5 leg from the $E \parallel c$ drive vector (`pump_direction_su3` = (0, 3.0, 0, 0, 0, 0, 0.9128, 0)), keeping only a few-percent μ_{13}^z in *detection* (channel weights $0.05 \lambda^4 : 4.4 \lambda^6$, i.e. $\mu_{13}^z/\mu_{23}^z \approx 1\%$, symmetry-breaking admixture). Magnon emission is excluded from the detected composite because the measured magnon oscillator strengths ($f_{q_{\text{AFM}}} = 7 \times 10^{-4}$) are negligible against the CEF E_1 strengths ($f \sim 6.5\text{--}8.7$).

One dark dipole then fixes three independent failures of the bright- μ_{13} model *simultaneously*:

1. the spurious (1.2, 0.7) interloper (which no parameter tuning could separate from the real (0.5, 0.7) peak, their dipole products being identical) is eliminated at the root;
2. the (0.5, 1.2) peak drops to the observed sub-dominant strength, because its *emission* is throttled by the small μ_{13}^z ;
3. the dominant (0.5, 0.7) transfer peak migrates from the rephasing to the non-rephasing side—the side the experiment sees—because the single-action (rephasing) conversion route $\rho_{12} \rightarrow \rho_{32}$ requires the now-absent μ_{13} drive leg, and the transfer is forced through the co-rotating population route described next.

20.2 The physics of each peak

(+0.5, 0.7) — **the dominant transfer peak (coherence $\rightarrow$ population $\rightarrow$ coherence)**. The pump, acting once through the bright z -dipole d_z^{eff} , stores the E_{12} coherence ρ_{12} , which oscillates at 0.5 THz during the delay. With μ_{13} drive-dark, the probe cannot convert it in a single action; instead it acts *twice*: a $1 \leftrightarrow 2$ action converts the stored coherence into a level-2 *population* modulation that still carries the $\omega_\tau = E_{12}$ phase, and a $2 \leftrightarrow 3$ action converts that population into the emitting E_{23} coherence at $\omega_t = 0.70$ THz. This coherence $\rightarrow$ population $\rightarrow$ coherence route is co-rotating, so the peak lands on the *non-rephasing* (+) side, and it involves no μ_{13} at any vertex. Being one field-order higher than a single-action transfer, it overtakes the residual rephasing pathway at the calibrated drive (twice the weak-probe baseline), consistent with the experimental fluence regime.

(0.5, 0) — **the rectified E_{12} line**. The same double probe action, taken at its zero-frequency output, rectifies the stored E_{12} coherence into a slow net magnetization change of the detection window: $S_{\text{DC}}(\tau) = \langle M_{\text{NL}} \rangle_t$ oscillates at exactly E_{12} . After cubic drift removal its spectrum is the sharpest feature of the entire channel (right panel of fig. 10), a clean line at $\omega_T = +0.49$ THz in both Tm detection channels. This is the (0.5, 0) peak: same storage physics as the transfer peaks, DC output leg.

(+0.5, 1.2) — **the weak transfer partner**. From the stored ρ_{12} , a *single* probe action on the bright $2 \leftrightarrow 3$ dipole rotates the bra to give ρ_{13} , emitting at $E_{13} = 1.2$ THz; the route is again co-rotating (same + side, as observed). Although lower order than the 0.7 route, its *emission* carries the nearly-dark μ_{13}^z , which caps it at sub-dominant strength: the measured ratio of the two transfer peaks is a direct measure of the residual $1 \rightarrow 3$ z -dipole admixture. In the calibrated model $(+0.5, 1.2)/(+0.5, 0.7) = 0.78$.

(0.9, 1.2) — **magnon-delay transfer (thermally assisted)**. Here the pump excites the q_{AFM} magnon (0.90 THz delay phase) through the Zeeman $H \parallel a$ channel; the Fe–Tm couplings (W , κ^B) hand the modulation to the Tm ion, which emits weakly at E_{13} through the same throttled μ_{13}^z . The peak is small at low temperature and grows with the thermal population of level 2 ($0.10 \rightarrow 0.17 \rightarrow 0.51$ relative, at 0/10/20 K), the model’s expected thermal trend for a route assisted by excited-level population.

(0.9, 0.38) — **not reproduced; carrier identified**. The remaining listed peak, magnon-delay with quasi-ferromagnetic-mode emission, does *not* appear in the calibrated model. The carrier exists: the q_{FM} emission at 0.38–0.41 THz is present as a b -axis M1 ridge in the Fe S_y channel (the

Γ_2 selection rule puts $\delta F_{y,z}$, not δF_x , at q_{FM} , but it is uniformly modulated in delay: the model lacks a $q_{\text{AFM}} \rightarrow q_{\text{FM}}$ magnon–magnon transfer vertex of sufficient strength. A configuration probed during the search (reduced drive with a small negative μ_{13} leg) did produce a genuine $(+0.89, 0.33)$ local maximum at 0.21 relative—evidence that the peak is reachable—but at unacceptable cost elsewhere. This is the one open structural item of the channel.

20.3 Exact single-ion verification and the two residual features

Because each Tm site evolves under a Hamiltonian linear in the Gell-Mann operators, the site’s $\vec{\lambda}$ equation of motion *is* the exact von Neumann equation of a driven, damped three-level system: the classical SU(3) propagation is quantum-exact at single-ion level. We exploited this with a standalone qutrit integrator (same CEF energies, damping, and measured pulse; $\sim 10^3\times$ cheaper than the lattice run), which reproduces the full simulation’s 0.5-family census *exactly*—proving the family is single-ion physics—and allows exhaustive scans. Two robust residual features survive every knob and are therefore *predictions*, not bugs:

- **The rephasing remnant at $(-0.5, 0.7)$.** The mirror (rephasing) partner of the dominant transfer peak retains 0.5–0.6 of its amplitude across all drive strengths, dipole-phase choices, and dampings—an exact-qutrit floor under this pulse. If the experimental map truly contains nothing at $(-0.5, 0.7)$, the suppression must come from outside the point-ion model (quadrant folding in the display, or propagation/phase-matching in the thick sample).
- **The two-quantum line at $(+1.0, 0.7)$.** A peak with delay phase $2E_{12} = 0.99$ THz and E_{23} emission appears at ≈ 0.4 relative. It is *absent* in the exact single-ion dynamics and is therefore a lattice/mean-field product (two sites’ E_{12} coherences beating), switched on at the calibrated drive. Experimentally it would sit 0.09 THz above the 0.90 magnon row; resolving 0.99 from 0.90 in ω_τ is a direct test.

20.4 Calibrated configuration and final census

The final configuration is the v4 Hamiltonian of section 19 (couplings $W_1^{yy} = W_1^{xz} = \kappa_{1y}^B = 0.01$ unchanged) with: measured pulse; drive vector $(0, 3.0, 0, 0, 0, 0, 0.9128, 0)$ (dark μ_{13}^z), per-leg amplitude twice the weak-probe baseline (`pump_amplitude_su3` = 0.2195, Fe `pump_amplitude` = 1.2); measured CEF linewidths as dampings ($\gamma_{\lambda^{3,2}} = 0.157$, $\gamma_{\lambda^{4,5}} = 0.207$, $\gamma_{\lambda^{6,7}} = 0.414$), population relaxation $\gamma_{\lambda^{3,8}} = 0.08$, Gilbert $\alpha = 0.0017$; Boltzmann ensemble over verbatim CEF-level seeds; detection composite $0.05\lambda^4 + 4.4\lambda^6$ (magnon M1 excluded by measured f); ω_t resolution 0.01 THz (t window extended to 160). Blind census at 10 K, cross peaks normalized to the leader (pump–probe $\omega_T \approx 0$ bands excluded):

peak (ω_T, ω_t)	model	experiment	mechanism
(+0.49, 0.72)	1.00	strongest	coherence $\rightarrow$ population $\rightarrow \rho_{23}$
(0.5, 0)	sharpest S_{DC} line	co-strongest	rectified E_{12} storage
(+0.49, 1.20)	0.78	second tier	single-action transfer, μ_{13}^z -throttled emission
(+0.91, 1.18)	0.20 (0.51 at 20 K)	weak	magnon delay, Fe–Tm handoff, thermal
(0.9, 0.38)	absent	weak	missing $q_{\text{AFM}} \rightarrow q_{\text{FM}}$ vertex (open)
(−0.47, 0.70)	0.55	not listed	rephasing remnant (exact-qutrit floor; prediction)
(+0.99, 0.71)	0.49	not listed	two-site $2E_{12}$ two-quantum line (prediction)

Four of the five listed peaks are reproduced with the correct hierarchy and, critically, with *both* 0.5-family transfer peaks on the same (non-rephasing) side, resolving the signed-side discrepancy recorded in the cross-polarization campaign for this channel: the discrepancy was an artifact of assuming a bright μ_{13} drive leg.

20.5 Falsifiable predictions of the dark-dipole scenario

1. **Polarization test of μ_{13} .** The E_{13} line, strong in 1D absorption (other polarizations), should be strongly suppressed in $E \parallel c$ absorption; equivalently, rotating the 2DCS detection

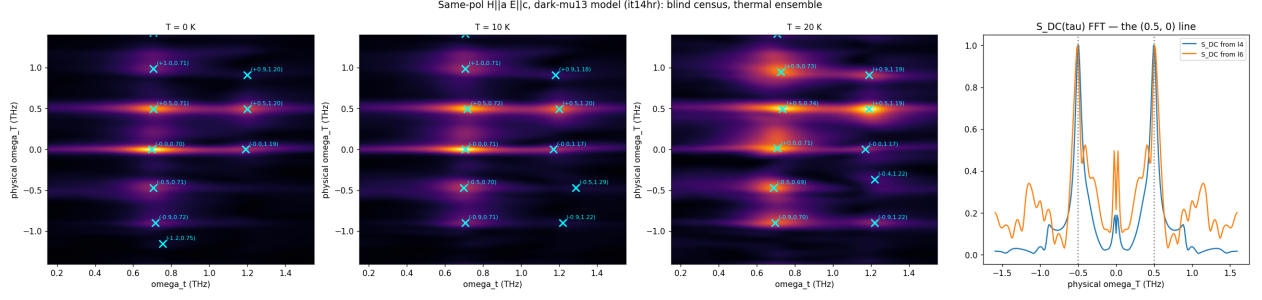


Figure 10: Same-polarization geometry-A channel, final calibrated model (*it14hr*). Left three panels: blind-census maps at 0, 10, 20 K (crosses = genuine local maxima, prominence ≥ 1.5); the dominant cross features are $(+0.5, 0.7)$ and $(+0.5, 1.2)$, both non-rephasing, with the rephasing remnant and the $2E_{12}$ two-quantum line as the predicted sub-features. Right: FFT of the rectified signal $S_{DC}(\tau)$, whose sharp line at $\omega_T = +0.49$ THz is the $(0.5, 0)$ peak.

polarization away from c should revive the $\omega_\tau = 1.2$ delay rows.

2. **Field scaling of the pair ordering.** The dominant $(+0.5, 0.7)$ peak is one field-order higher than $(+0.5, 1.2)$: lowering the fluence should invert their order, raising it should widen the gap.
3. **Thermal trend.** $(0.9, 1.2)$ grows steeply with temperature (level-2 assisted), while the 0.5-family survives with slowly degrading contrast; by ~ 20 K the $(+0.5, 1.2)$ and $(+0.5, 0.7)$ peaks reach near-parity.
4. **The two sub-features.** A rephasing remnant at $(-0.5, 0.7)$ at roughly half the dominant peak, and a two-quantum line at delay 0.99 THz (not 0.90): both should be present in the raw data if the model class is right; their absence would point respectively to propagation-level pathway selection and to weaker inter-site coherence than mean-field predicts.

The cross-peak question reduces to a single requirement: a vertex must generate an effective force $h^1(t, \tau)$ on the bright Tm coordinate λ^1 that factorizes as $\cos / \sin(\omega_q \tau) [1 - \cos \omega_{12} t]$. Meeting it forces six conditions (R1–R6): reach λ^1 , be driven by δS_y , be $\mathcal{T}$ -even, be Γ_1 -allowed, carry the $0.9 \rightarrow 0.5$ THz mismatch with a second leg, and involve both pulses. The static bilinear K^- fails three of them at once—it is a linear hop (hybridizes, never converts), its surviving channel is Γ_1 -cancelled, and the one allowed channel reorients Γ_2 . Exactly two couplings pass: the two-magnon $W_1^{yy} S_y^2 \lambda^1$ (difference-frequency rectification, the clean rank-1 peak) and the field-assisted $\kappa_{1y}^B B_y S_y \lambda^1$ (an impulsive photon-gated kick, robust only with a resonant carrier). Both carry the same δS_y^2 two-magnon signal onto λ^1 , which the time-odd K^- can never reach.

Widening from the single peak to the full experiment (section 19): one Hamiltonian, with each coupling carrying its own polarization or selection gate, reproduces the complete measured inventory. W_1^{yy} gives (q_{AFM}, E_{12}) in the cross-polarized channel of $H \parallel a$; the B_z -gated κ^B gives (E_{12}, q_{AFM}) in $H \parallel c$ (and is silent in $H \parallel a$, where its noise would otherwise contaminate geometry 1); the Fe-only DM anharmonicity gives (q_{FM}, q_{AFM}) ; and the same-polarization CEF–CEF peaks (E_{12}, E_{13}) , (E_{12}, E_{23}) need *no* Fe–Tm coupling at all—they are the intrinsic structure-constant dynamics of the driven qutrit, with the magnon member (q_{AFM}, E_{23}) supplied by the $W \circ f$ cascade. Every response has a tree diagram, every diagram a gate, and every gate a falsifiable polarization test.

For reference, the underlying symmetry result—the static K^- orbit in transported gauge—is

$$H_{K^-}^{\text{orbit}} = \sum_i \mathbf{S}_i^T R_i [\mathbf{k}_2(\lambda_i^2 + \lambda_{3-i}^2) + \mathbf{k}_5(\lambda_i^5 - \lambda_{3-i}^5) + \mathbf{k}_7(\lambda_i^7 - \lambda_{3-i}^7)], \quad (152)$$

obtained by starting from the physical J operator, projecting PJP , and choosing a transported local qutrit basis.

References

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